https://slideplayer.com/slide/2804957/

(VVVVVVVVVVVVVVVVVV IMPT APPLICATION)

<u>Diagonalization of a Matrix:</u>

$$\mathbf{D = P^{-1}AP = \begin{bmatrix} \lambda_1 & 0 & 0 & - & 0 \\ 0 & \lambda_2 & 0 & - & 0 \\ 0 & 0 & \lambda_3 & - & 0 \\ - & - & - & - & - \\ 0 & 0 & 0 & 0 & \lambda_n \end{bmatrix}}$$

Where,
D is a diagonalized matrix consisting of Eigen values.
P is a matrix consisting of Eigen vectors
$\mathbf{P^{-1}}$ is a inverse matrix of P
A is given Matrix
$\lambda_1, \lambda_2, \lambda_3, \ldots\ldots\ldots\ldots\ldots\ldots, \lambda_n$ are eigen values of A.

Example 49: Diagonalize the matrix $\begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$

Solution:

Let, $\mathbf{A} = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$

$$\Rightarrow \mathbf{A} - \lambda \mathbf{I} = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \mathbf{A} - \lambda \mathbf{I} = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$\Rightarrow \mathbf{A} - \lambda \mathbf{I} = \begin{bmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{bmatrix}$$

$$|\mathbf{A} - \lambda \mathbf{I}| = \begin{vmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{vmatrix}$$

$$= (1-\lambda)[(-5-\lambda)(4-\lambda) - (-6)(3)] - (-3)[3(4-\lambda) - 3\times 6] + 3[3(-6) - 6(-5-\lambda)]$$

$$= (1-\lambda)[(-20+5\lambda-4\lambda+\lambda^2)+18]+3[12-3\lambda-18]+3[-18+30+6\lambda]$$
$$= (1-\lambda)[\lambda+\lambda^2-2]+3[-3\lambda-6]+3[12+6\lambda]$$
$$= \lambda+\lambda^2-2-\lambda^2-\lambda^3+2\lambda-9\lambda-18+36+18\lambda$$
$$-\lambda^3+12\lambda+16$$

Now,

$$|\mathbf{A}-\lambda\mathbf{I}| = 0$$
$$\Rightarrow -\lambda^3+12\lambda+16 = 0$$
$$\Rightarrow \lambda^3-12\lambda-16 = 0$$
$$\Rightarrow \lambda^3+2\lambda^2-2\lambda^2-4\lambda-8\lambda-16 = 0$$
$$\Rightarrow \lambda^2(\lambda+2)-2\lambda(\lambda+2)-8(\lambda+2) = 0$$
$$\Rightarrow (\lambda+2)(\lambda^2-2\lambda-8) = 0$$
$$\Rightarrow (\lambda+2)(\lambda^2-4\lambda+2\lambda-8) = 0$$
$$\Rightarrow (\lambda+2)(\lambda-4)(\lambda+2) = 0$$
$$\Rightarrow \lambda = -2,\ 4,\ -2$$

Let, $\mathbf{v_1} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

We have, $\mathbf{Av_1} = \lambda\mathbf{v_1}$

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \lambda\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = 4\begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad [\lambda = 4]$$

$$\Rightarrow \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4x \\ 4y \\ 4z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x-3y+3z \\ 3x-5y+3z \\ 6x-6y+4z \end{bmatrix} = \begin{bmatrix} 4x \\ 4y \\ 4z \end{bmatrix}$$

$$\Rightarrow \quad x-3y+3z = 4x$$
$$3x-5y+3z = 4y$$
$$6x-6y+4z = 4z$$
$$\Rightarrow \quad -3x-3y+3z = 0$$

$$\begin{aligned} 3x - 9y + 3z &= 0 \\ 6x - 6y &= 0 \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} x + y - z &= 0 \\ x - 3y + z &= 0 \\ x - y &= 0 \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} x + y - z &= 0 \\ x - 3y + z &= 0 \\ x - y + 0.z &= 0 \end{aligned}$$

We have,

$\mathbf{L_i \to -a_{i1}L_1 + a_{11}L_i}$

$\mathbf{i = 2,\ L_2 \to -a_{21}L_1 + a_{11}L_2}$

$$= -1(x + y - z) + 1(x - 3y + z)$$

$$= -4y + 2z$$

$\mathbf{i = 3,\ L_3 \to -a_{31}L_1 + a_{11}L_3}$

$L_3 \to -1(x + y - z) + 1(x - y + 0.z)$

$$= -2y + z$$

Thus we obtain following system

$$\begin{aligned} x + y - z &= 0 \\ -4y + 2z &= 0 \\ -2y + z &= 0 \end{aligned}$$

$\Rightarrow x + y - z = 0$

$2y - z = 0$ [Dividing by -2 on both sides]

$2y - z = 0$ [Multiplying by -1 on both sides]

Since the 2nd and 3rd equations are identical. We can disregard one of them

$$\Rightarrow \begin{aligned} x + y - z &= 0 \\ 2y - z &= 0 \end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular (3-2) =1 free variable, which is z.
[Note Here total variables are three, i.e x, y, z and blocked variable is x and y, hence free variable is z]

Let, $z = 2$

$\therefore y = 1$ and $x = 1$

$$\therefore\ \mathbf{v_1} = \begin{pmatrix} \mathbf{1} \\ \mathbf{1} \\ \mathbf{2} \end{pmatrix}$$

$$\text{Let, } \mathbf{v_2} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix}$$

We have,

$$\mathbf{Av_2 = \lambda v_2}$$

$$\begin{bmatrix} \mathbf{1} & \mathbf{-3} & \mathbf{3} \\ \mathbf{3} & \mathbf{-5} & \mathbf{3} \\ \mathbf{6} & \mathbf{-6} & \mathbf{4} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix} = \lambda \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \mathbf{1} & \mathbf{-3} & \mathbf{3} \\ \mathbf{3} & \mathbf{-5} & \mathbf{3} \\ \mathbf{6} & \mathbf{-6} & \mathbf{4} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix} = \mathbf{-2} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix} \quad [\lambda = \mathbf{-2}]$$

$$\Rightarrow \begin{bmatrix} \mathbf{1} & \mathbf{-3} & \mathbf{3} \\ \mathbf{3} & \mathbf{-5} & \mathbf{3} \\ \mathbf{6} & \mathbf{-6} & \mathbf{4} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix} = \begin{bmatrix} -2\mathbf{x} \\ -2\mathbf{y} \\ -2\mathbf{z} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \mathbf{x - 3y + 3z} \\ \mathbf{3x - 5y + 3z} \\ \mathbf{6x - 6y + 4z} \end{bmatrix} = \begin{bmatrix} -2\mathbf{x} \\ -2\mathbf{y} \\ -2\mathbf{z} \end{bmatrix}$$

$$\Rightarrow \begin{aligned} &\mathbf{x - 3y + 3z = -2x} \\ &\mathbf{3x - 5y + 3z = -2y} \\ &\mathbf{6x - 6y + 4z = -2z} \end{aligned}$$

$$\Rightarrow \begin{aligned} &3\mathbf{x - 3y + 3z = 0} \\ &\mathbf{3x - 3y + 3z = 0} \\ &\mathbf{6x - 6y + 6z = 0} \end{aligned}$$

$$\Rightarrow \begin{aligned} &\mathbf{x - y + z = 0} \\ &\mathbf{x - y + z = 0} \\ &\mathbf{x - y + z = 0} \end{aligned}$$

Since the 1st, 2nd and 3rd equations are identical. We can disregard two of them

$$\Rightarrow \quad \mathbf{x - y + z = 0}$$

In echelon form, there are only one equation in three unknowns, then the system has a non-zero solution and in particular (3-1) =2 free variable, which is y and z.

Let, y = 1 and z =0

$\therefore \quad x = 1$

$$\therefore \mathbf{v_2} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix} = \begin{pmatrix} \mathbf{1} \\ \mathbf{1} \\ \mathbf{0} \end{pmatrix}$$

$$\text{Let, } \mathbf{v_3} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix}$$

We have,

$$\mathbf{Av_3} = \lambda \mathbf{v_3}$$

$$\begin{bmatrix} \mathbf{1} & \mathbf{-3} & \mathbf{3} \\ \mathbf{3} & \mathbf{-5} & \mathbf{3} \\ \mathbf{6} & \mathbf{-6} & \mathbf{4} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix} = \lambda \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \mathbf{1} & \mathbf{-3} & \mathbf{3} \\ \mathbf{3} & \mathbf{-5} & \mathbf{3} \\ \mathbf{6} & \mathbf{-6} & \mathbf{4} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix} = \mathbf{-2} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix} \quad [\lambda = \mathbf{-2}]$$

$$\Rightarrow \begin{bmatrix} \mathbf{1} & \mathbf{-3} & \mathbf{3} \\ \mathbf{3} & \mathbf{-5} & \mathbf{3} \\ \mathbf{6} & \mathbf{-6} & \mathbf{4} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix} = \begin{bmatrix} -2\mathbf{x} \\ -2\mathbf{y} \\ -2\mathbf{z} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \mathbf{x - 3y + 3z} \\ \mathbf{3x - 5y + 3z} \\ \mathbf{6x - 6y + 4z} \end{bmatrix} = \begin{bmatrix} -2\mathbf{x} \\ -2\mathbf{y} \\ -2\mathbf{z} \end{bmatrix}$$

$$\Rightarrow \quad \begin{aligned} &\mathbf{x - 3y + 3z = -2x} \\ &\mathbf{3x - 5y + 3z = -2y} \\ &\mathbf{6x - 6y + 4z = -2z} \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} &3\mathbf{x - 3y + 3z = 0} \\ &\mathbf{3x - 3y + 3z = 0} \\ &\mathbf{6x - 6y + 6z = 0} \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} &\mathbf{x - y + z = 0} && \text{[Dividing by 3 on both sides]} \\ &\mathbf{x - y + z = 0} && \text{[Dividing by 3 on both sides]} \\ &\mathbf{x - y + z = 0} && \text{[Dividing by 6 on both sides]} \end{aligned}$$

Since the 1st, 2nd and 3rd equations are identical. We can disregard two of them

$$\Rightarrow \quad \mathbf{x - y + z = 0}$$

In echelon form, there are only one equation in three unknowns, then the system has a non-zero solution and in particular (3-1) =2 free variable, which is y and z.

Let, y = 0 and z =1

$\therefore \quad x = -1$

$$\therefore \mathbf{v_3} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix} = \begin{pmatrix} \mathbf{-1} \\ \mathbf{0} \\ \mathbf{1} \end{pmatrix}$$

$$\mathbf{P} = [\mathbf{v_1} \quad \mathbf{v_2} \quad \mathbf{v_3}]$$

$$\mathbf{P} = \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{-1} \\ \mathbf{1} & \mathbf{1} & \mathbf{0} \\ \mathbf{2} & \mathbf{0} & \mathbf{1} \end{bmatrix}$$

We know that, $\mathbf{D = P^{-1}AP}$

Now,

$$\mathbf{P} = \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{-1} \\ \mathbf{1} & \mathbf{1} & \mathbf{0} \\ \mathbf{2} & \mathbf{0} & \mathbf{1} \end{bmatrix}$$

$\therefore$ Co-factor of $\mathbf{a_{11}(1) = c_{11} = +\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1}$

$\therefore$ Co-factor of $\mathbf{a_{12}(1) = c_{12} = -\begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = -1}$

$\therefore$ Co-factor of $\mathbf{a_{13}(-1) = c_{13} = +\begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} = -2}$

$\therefore$ Co-factor of $\mathbf{a_{21}(1) = c_{21} = -\begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = -1}$

$\therefore$ Co-factor of $\mathbf{a_{22}(1) = c_{22} = +\begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = 3}$

$\therefore$ Co-factor of $\mathbf{a_{23}(0) = c_{23} = -\begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} = 2}$

$\therefore$ Co-factor of $\mathbf{a_{31}(2) = c_{31} = +\begin{vmatrix} 1 & -1 \\ 1 & 0 \end{vmatrix} = 1}$

$\therefore$ Co-factor of $\mathbf{a_{32}(0) = c_{32}} = -\begin{vmatrix} \mathbf{1} & \mathbf{-1} \\ \mathbf{1} & \mathbf{0} \end{vmatrix} = \mathbf{-1}$

$\therefore$ Co-factor of $\mathbf{a_{33}(1) = c_{33}} = +\begin{vmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} \end{vmatrix} = \mathbf{0}$

Co-factor Matrix: $\mathbf{P^C} = \begin{bmatrix} \mathbf{1} & \mathbf{-1} & \mathbf{-2} \\ \mathbf{-1} & \mathbf{3} & \mathbf{2} \\ \mathbf{1} & \mathbf{-1} & \mathbf{0} \end{bmatrix}$

Adjoint of $\mathrm{P} = (\mathrm{P^C})^{\mathrm{T}} = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ -2 & 2 & 0 \end{bmatrix}$

Given, $\mathbf{P} = \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{-1} \\ \mathbf{1} & \mathbf{1} & \mathbf{0} \\ \mathbf{2} & \mathbf{0} & \mathbf{1} \end{bmatrix}$

$$|\mathbf{P}| = \begin{vmatrix} \mathbf{1} & \mathbf{1} & \mathbf{-1} \\ \mathbf{1} & \mathbf{1} & \mathbf{0} \\ \mathbf{2} & \mathbf{0} & \mathbf{1} \end{vmatrix} = \mathbf{1(1 \times 1 - 0) - 1(1 - 0) + (-1)(0 - 2) = 1 - 1 + 2 = 2}$$

$$\therefore \mathbf{P^{-1}} = \frac{\mathbf{Adjoint\ of\ P}}{|\mathbf{P}|} = \frac{\mathbf{1}}{\mathbf{2}} \begin{bmatrix} \mathbf{1} & \mathbf{-1} & \mathbf{1} \\ \mathbf{-1} & \mathbf{3} & \mathbf{-1} \\ \mathbf{-2} & \mathbf{2} & \mathbf{0} \end{bmatrix}$$

$$\boldsymbol{D = P^{-1}AP} = \frac{\mathbf{1}}{\mathbf{2}} \begin{bmatrix} \mathbf{1} & \mathbf{-1} & \mathbf{1} \\ \mathbf{-1} & \mathbf{3} & \mathbf{-1} \\ \mathbf{-2} & \mathbf{2} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{1} & \mathbf{-3} & \mathbf{3} \\ \mathbf{3} & \mathbf{-5} & \mathbf{3} \\ \mathbf{6} & \mathbf{-6} & \mathbf{4} \end{bmatrix} \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{-1} \\ \mathbf{1} & \mathbf{1} & \mathbf{0} \\ \mathbf{2} & \mathbf{0} & \mathbf{1} \end{bmatrix}$$

$$= \frac{\mathbf{1}}{\mathbf{2}} \begin{bmatrix} \mathbf{1} & \mathbf{-1} & \mathbf{1} \\ \mathbf{-1} & \mathbf{3} & \mathbf{-1} \\ \mathbf{-2} & \mathbf{2} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{1 \times 1 - 3 \times 1 + 3 \times 2} & \mathbf{1 \times 1 - 3 \times 1 + 3 \times 0} & \mathbf{1(-1) + (-3)(0) + 3 \times 1} \\ \mathbf{3 \times 1 - 5 \times 1 + 3 \times 2} & \mathbf{3 \times 1 - 5 \times 1 + 3 \times 0} & \mathbf{3(-1) - 5(0) + 3 \times 1} \\ \mathbf{6 \times 1 - 6 \times 1 + 4 \times 2} & \mathbf{6 \times 1 - 6 \times 1 + 4 \times 0} & \mathbf{6(-1) - 6 \times 0 + 4 \times 1} \end{bmatrix}$$

$$= \frac{\mathbf{1}}{\mathbf{2}} \begin{bmatrix} \mathbf{1} & \mathbf{-1} & \mathbf{1} \\ \mathbf{-1} & \mathbf{3} & \mathbf{-1} \\ \mathbf{-2} & \mathbf{2} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{1 - 3 + 6} & \mathbf{1 - 3 + 0} & \mathbf{-1 + 0 + 3} \\ \mathbf{3 - 5 + 6} & \mathbf{3 - 5 + 0} & \mathbf{-3 - 0 + 3} \\ \mathbf{6 - 6 + 8} & \mathbf{6 - 6 + 0} & \mathbf{-6 - 0 + 4} \end{bmatrix}$$

$$=\frac{1}{2}\begin{bmatrix}1 & -1 & 1\\ -1 & 3 & -1\\ -2 & 2 & 0\end{bmatrix}\begin{bmatrix}4 & -2 & 2\\ 4 & -2 & 0\\ 8 & 0 & -2\end{bmatrix}$$

$$=\frac{1}{2}\begin{bmatrix}1\times 4-1\times 4+1\times 8 & 1(-2)+(-1)(-2)+1\times 0 & 1\times 2-1\times 0+1(-2)\\ -1\times 4+3\times 4+(-1)\times 8 & (-1)(-2)+3(-2)+(-1)\times 0 & (-1)\times 2+3\times 0+(-1)(-2)\\ -2\times 4+2\times 4+0\times 8 & (-2)(-2)+2(-2)+0\times 0 & -2\times 2+2\times 0+0(-2)\end{bmatrix}$$

$$=\frac{1}{2}\begin{bmatrix}4-4+8 & -2+2 & 2-2\\ -4+12+-8 & 2-6 & -2+2\\ -8+8+0 & 4-4 & -4)\end{bmatrix}$$

$$=\frac{1}{2}\begin{bmatrix}8 & 0 & 0\\ 0 & -4 & 0\\ 0 & 0 & -4\end{bmatrix}$$

$$D=\begin{bmatrix}4 & 0 & 0\\ 0 & -2 & 0\\ 0 & 0 & -2\end{bmatrix}$$

Example 50: Diagonalize the matrix: $\begin{bmatrix}4 & 1\\ 1 & 4\end{bmatrix}$

Solution:

Let, $\mathbf{A}=\begin{bmatrix}4 & 1\\ 1 & 4\end{bmatrix}$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{bmatrix}4 & 1\\ 1 & 4\end{bmatrix}-\lambda\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{bmatrix}4 & 1\\ 1 & 4\end{bmatrix}-\begin{bmatrix}\lambda & 0\\ 0 & \lambda\end{bmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{bmatrix}4-\lambda & 1\\ 1 & 4-\lambda\end{bmatrix}$$

$$|\mathbf{A}-\lambda\mathbf{I}|=\begin{vmatrix}4-\lambda & 1\\ 1 & 4-\lambda\end{vmatrix}$$

$$|\mathbf{A}-\lambda\mathbf{I}|=(4-\lambda)(4-\lambda)-1$$

$$|\mathbf{A}-\lambda\mathbf{I}|=16-4\lambda-4\lambda+\lambda^2-1$$

$$|\mathbf{A}-\lambda\mathbf{I}|=\lambda^2-8\lambda+15$$

Now,

$$|\mathbf{A} - \lambda\mathbf{I}| = 0$$
$$\Rightarrow |\mathbf{A} - \lambda\mathbf{I}| = \lambda^2 - 8\lambda + 15 = 0$$
$$\Rightarrow |\mathbf{A} - \lambda\mathbf{I}| = \lambda^2 - 3\lambda - 5\lambda + 15 = 0$$
$$\Rightarrow |\mathbf{A} - \lambda\mathbf{I}| = \lambda^2 - 3\lambda - 5\lambda + 15 = 0$$
$$\Rightarrow |\mathbf{A} - \lambda\mathbf{I}| = (\lambda - 3)(\lambda - 5) = 0$$
$$\Rightarrow (\lambda - 3)(\lambda - 5) = 0$$
$$\Rightarrow \lambda = 3, \lambda = 5$$

We have, $\mathbf{Av_1} = \lambda\mathbf{v_1}$

Let, $\mathbf{v_1} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$

$$\mathbf{Av_1} = \lambda\mathbf{v_1}$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \lambda\begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 3\begin{bmatrix} x \\ y \end{bmatrix} \text{ [When, } \lambda = 3 \text{]}$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \end{bmatrix}$$

$$\begin{bmatrix} 4 \times x + 1 \times y \\ 1 \times x + 4 \times y \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \end{bmatrix}$$

$$\begin{bmatrix} 4x + y \\ x + 4y \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \end{bmatrix}$$

$$\Rightarrow \quad \begin{aligned} 4x + y &= 3x \\ x + 4y &= 3y \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} x + y &= 0 \\ x + y &= 0 \end{aligned}$$

Since the 1st and 2nd equations are identical. We can disregard one of them

$$\Rightarrow \quad x + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Let $y = 1$, $x =$ -1

$$\therefore \mathbf{v_1} = \begin{pmatrix} \mathbf{-1} \\ \mathbf{1} \end{pmatrix}$$

We have, $\mathbf{Av_2} = \lambda \mathbf{v_2}$

Let, $\mathbf{v_2} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$

$$\begin{bmatrix} \mathbf{4} & \mathbf{1} \\ \mathbf{1} & \mathbf{4} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \lambda \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{4} & \mathbf{1} \\ \mathbf{1} & \mathbf{4} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \mathbf{5} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} \text{ [When, } \lambda = \mathbf{5}\text{]}$$

$$\begin{bmatrix} \mathbf{4} & \mathbf{1} \\ \mathbf{1} & \mathbf{4} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{5x} \\ \mathbf{5y} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{4 \times x + 1 \times y} \\ \mathbf{1 \times x + 4 \times y} \end{bmatrix} = \begin{bmatrix} \mathbf{5x} \\ \mathbf{5y} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{4x + y} \\ \mathbf{x + 4y} \end{bmatrix} = \begin{bmatrix} \mathbf{5x} \\ \mathbf{5y} \end{bmatrix}$$

$$\Rightarrow \quad \begin{aligned} \mathbf{4x + y} &= \mathbf{5x} \\ \mathbf{x + 4y} &= \mathbf{5y} \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} \mathbf{-x + y} &= \mathbf{0} \\ \mathbf{x - y} &= \mathbf{0} \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} \mathbf{x - y} &= \mathbf{0} \\ \mathbf{x - y} &= \mathbf{0} \end{aligned} \qquad \text{[Multiplying by -1 on both sides]}$$

Since the 1stand 2nd equations are identical. We can disregard one of them

$$\Rightarrow \quad \mathbf{x - y = 0}$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Let, $y = 1 \quad \therefore x = 1$

$$\mathbf{v_2} = \begin{pmatrix} \mathbf{1} \\ \mathbf{1} \end{pmatrix}$$

$$P = [v_1 \quad v_2]$$

$$\therefore P = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$|\mathbf{P}| = (-1) \times 1 - 1 = -2$$

We know that, $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$

Now,

$$\therefore \mathbf{P} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$\therefore$ Co-factor of $\mathbf{a_{11}(-1) = c_{11} = +1}$

$\therefore$ Co-factor of $\mathbf{a_{12}(1) = c_{12} = -1}$

$\therefore$ Co-factor of $\mathbf{a_{21}(1) = c_{21} = -1}$

$\therefore$ Co-factor of $\mathbf{a_{22}(1) = c_{22} = +(-1) = -1}$

Co-factor Matrix: $\mathbf{P^C} = \begin{bmatrix} \mathbf{1} & \mathbf{-1} \\ \mathbf{-1} & \mathbf{-1} \end{bmatrix}$

Adjoint of $P = (P^C)^T = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$

$$\therefore P^{-1} = \frac{\text{Adjoint of P}}{|P|} = \frac{1}{-2}\begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

$$\boldsymbol{D = P^{-1}AP} = \frac{\mathbf{1}}{\mathbf{-2}}\begin{bmatrix} \mathbf{1} & \mathbf{-1} \\ \mathbf{-1} & \mathbf{-1} \end{bmatrix}\begin{bmatrix} \mathbf{4} & \mathbf{1} \\ \mathbf{1} & \mathbf{4} \end{bmatrix}\begin{bmatrix} \mathbf{-1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} \end{bmatrix}$$

$$\boldsymbol{D = P^{-1}AP} = \frac{\mathbf{1}}{\mathbf{-2}}\begin{bmatrix} \mathbf{1} & \mathbf{-1} \\ \mathbf{-1} & \mathbf{-1} \end{bmatrix}\begin{bmatrix} \mathbf{4(-1) + 1 \times 1} & \mathbf{4 \times 1 + 1 \times 1} \\ \mathbf{1(-1) + 4 \times 1} & \mathbf{1 \times 1 + 4 \times 1} \end{bmatrix}$$

$$\boldsymbol{D = P^{-1}AP} = \frac{\mathbf{1}}{\mathbf{-2}}\begin{bmatrix} \mathbf{1} & \mathbf{-1} \\ \mathbf{-1} & \mathbf{-1} \end{bmatrix}\begin{bmatrix} \mathbf{-4+1} & \mathbf{4+1} \\ \mathbf{-1+4} & \mathbf{1+4} \end{bmatrix}$$

$$\boldsymbol{D = P^{-1}AP} = \frac{\mathbf{1}}{\mathbf{-2}}\begin{bmatrix} \mathbf{1} & \mathbf{-1} \\ \mathbf{-1} & \mathbf{-1} \end{bmatrix}\begin{bmatrix} \mathbf{-3} & \mathbf{5} \\ \mathbf{3} & \mathbf{5} \end{bmatrix}$$

$$\boldsymbol{D = P^{-1}AP} = \frac{\mathbf{1}}{\mathbf{-2}}\begin{bmatrix} \mathbf{1(-3) + (-1) \times 3} & \mathbf{1 \times 5 + (-1) \times 5} \\ \mathbf{(-1)(-3) + (-1) \times 3} & \mathbf{(-1) \times 5 + (-1) \times 5} \end{bmatrix}$$

$$\boldsymbol{D = P^{-1}AP} = \frac{\mathbf{1}}{\mathbf{-2}}\begin{bmatrix} \mathbf{-3-3} & \mathbf{5-5} \\ \mathbf{3-3} & \mathbf{-5-5} \end{bmatrix}$$

$$\boldsymbol{D = P^{-1}AP} = \frac{\mathbf{1}}{\mathbf{-2}}\begin{bmatrix} \mathbf{-6} & \mathbf{0} \\ \mathbf{0} & \mathbf{-10} \end{bmatrix}$$

$$D = P^{-1}AP = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$$

$$\therefore \mathbf{D} = \mathbf{P}^{-1}\mathbf{AP} = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix} \quad \textit{Answer}$$

Example 51: **Diagonalize the matrix:** $\begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix}$

Solution:

Given, $\mathbf{A} = \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix}$

For Eigen values,

$$\Rightarrow \mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} - \lambda\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$\Rightarrow \mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} 5-\lambda & 1 & 2 \\ 0 & 3-\lambda & 0 \\ 2 & 1 & 5-\lambda \end{bmatrix}$$

$$\Rightarrow |\mathbf{A} - \lambda\mathbf{I}| = \begin{vmatrix} 5-\lambda & 1 & 2 \\ 0 & 3-\lambda & 0 \\ 2 & 1 & 5-\lambda \end{vmatrix}$$

$$= (5-\lambda)[(3-\lambda)(5-\lambda) - 0] - 1[0(5-\lambda) - 2\times 0] + 2[0\times 1 - 2(3-\lambda)]$$

$$= (5-\lambda)[15 - 3\lambda - 5\lambda + \lambda^2] - 1[0-0] + 2[0 - 6 + 2\lambda]$$

$$= (5-\lambda)[15 - 8\lambda + \lambda^2] + 2[-6 + 2\lambda]$$

$$= (5-\lambda)[15 - 8\lambda + \lambda^2] - 12 + 4\lambda$$

$$= 75 - 40\lambda + 5\lambda^2 - 15\lambda + 8\lambda^2 - \lambda^3 - 12 + 4\lambda$$

$$= 63 - 51\lambda + 13\lambda^2 - \lambda^3$$

$$= -\lambda^3 + 13\lambda^2 - 51\lambda + 63$$

$$\therefore |\mathbf{A} - \lambda\mathbf{I}| = 0$$

$$\Rightarrow -\lambda^3 + 13\lambda^2 - 51\lambda + 63 = 0$$

$$\Rightarrow \lambda^3 - 13\lambda^2 + 51\lambda - 63 = 0$$

$$\Rightarrow \lambda^3 - 3\lambda^2 - 10\lambda^2 + 30\lambda + 21\lambda - 63 = 0$$

$$\Rightarrow \lambda^2(\lambda - 3) - 10\lambda(\lambda - 3) + 21(\lambda - 3) = 0$$

$$\Rightarrow (\lambda - 3)(\lambda^2 - 10\lambda + 21) = 0$$

$$\Rightarrow (\lambda - 3)(\lambda^2 - 7\lambda - 3\lambda + 21) = 0$$

$$\Rightarrow (\lambda - 3)[\lambda(\lambda - 7) - 3(\lambda - 7)] = 0$$

$$\Rightarrow (\lambda - 3)(\lambda - 7)(\lambda - 3) = 0$$

$$\Rightarrow \lambda = 3, \lambda = 7, \lambda = 3$$
$$\Rightarrow \lambda = 3, \lambda = 3, \lambda = 7$$

Let a nonzero vector $\mathbf{v_1} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix}$

We have, $\mathbf{Av_1} = \lambda \mathbf{v_1}$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 3 \begin{bmatrix} x \\ y \\ z \end{bmatrix} [\text{When}, \lambda = 3]$$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \\ 3z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5x + y + 2z \\ 0 \times x + 3y + 0 \times z \\ 2x + y + 5z \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \\ 3z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5x + y + 2z \\ 3y \\ 2x + y + 5z \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \\ 3z \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 5x + y + 2z &= 3x \\ 3y &= 3y \\ 2x + y + 5z &= 3z \end{aligned}$$

$$\Rightarrow \begin{aligned} 2x + y + 2z &= 0 \\ 3y - 3y &= 0 \\ 2x + y + 2z &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} 2x + y + 2z &= 0 \\ 0 &= 0 \\ 2x + y + 2z &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} 2x + y + 2z &= 0 \\ 2x + y + 2z &= 0 \end{aligned}$$

Since the 1st and 2nd equations are identical. We can disregard one of them

$$\Rightarrow 2x + y + 2z = 0$$

In echelon form, there are only one equation in three unknowns, then the system has a non-zero solution and in particular (3-1) =2 free variable, which is y and z.

Let,

$z = 1$

$y = 2$

$\therefore 2x + 2 + 2 = 0$

$\Rightarrow x = -2$

$$\therefore \mathbf{v_1} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix} \text{ For } \lambda = 3$$

Let, $\mathbf{v_2} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

We have, $\mathbf{Av_2} = \lambda \mathbf{v_2}$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 3 \begin{bmatrix} x \\ y \\ z \end{bmatrix} [\text{When}, \lambda = 3]$$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \\ 3z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5x + y + 2z \\ 0 \times x + 3y + 0 \times z \\ 2x + y + 5z \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \\ 3z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \mathbf{5x + y + 2z} \\ \mathbf{3y} \\ \mathbf{2x + y + 5z} \end{bmatrix} = \begin{bmatrix} 3\mathbf{x} \\ 3\mathbf{y} \\ 3\mathbf{z} \end{bmatrix}$$

$$\Rightarrow \quad \begin{aligned} \mathbf{5x + y + 2z} &= \mathbf{3x} \\ \mathbf{3y} &= \mathbf{3y} \\ \mathbf{2x + y + 5z} &= \mathbf{3z} \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} \mathbf{2x + y + 2z} &= \mathbf{0} \\ \mathbf{3y - 3y} &= \mathbf{0} \\ \mathbf{2x + y + 2z} &= \mathbf{0} \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} \mathbf{2x + y + 2z} &= \mathbf{0} \\ \mathbf{0} &= \mathbf{0} \\ \mathbf{2x + y + 2z} &= \mathbf{0} \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} \mathbf{2x + y + 2z} &= \mathbf{0} \\ \mathbf{2x + y + 2z} &= \mathbf{0} \end{aligned}$$

Since the 1stand 2nd equations are identical. We can disregard one of them

$$\Rightarrow \quad \mathbf{2x + y + 2z = 0}$$

In echelon form, there are only one equation in three unknowns, then the system has a non-zero solution and in particular (3-1) =2 free variable, which is y and z.

Let,

$z = 1$

$y = 0$

$\therefore 2x + 0 + 2 = 0$

$\Rightarrow x = -1$

$$\therefore \mathbf{v_2} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix} = \begin{bmatrix} \mathbf{-1} \\ \mathbf{0} \\ \mathbf{1} \end{bmatrix}$$

Let, $\mathbf{v_3} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix}$

We have, $\mathbf{Av_3} = \lambda \mathbf{v_3}$

$$\Rightarrow \begin{bmatrix} \mathbf{5} & \mathbf{1} & \mathbf{2} \\ \mathbf{0} & \mathbf{3} & \mathbf{0} \\ \mathbf{2} & \mathbf{1} & \mathbf{5} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix} = \lambda \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 7 \begin{bmatrix} x \\ y \\ z \end{bmatrix} [\text{When}, \lambda = 7]$$

$$\Rightarrow \begin{bmatrix} 5 & 1 & 2 \\ 0 & 3 & 0 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7x \\ 7y \\ 7z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5x + y + 2z \\ 0 \times x + 3y + 0 \times z \\ 2x + y + 5z \end{bmatrix} = \begin{bmatrix} 7x \\ 7y \\ 7z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5x + y + 2z \\ 3y \\ 2x + y + 5z \end{bmatrix} = \begin{bmatrix} 7x \\ 7y \\ 7z \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 5x + y + 2z &= 7x \\ 3y &= 7y \\ 2x + y + 5z &= 7z \end{aligned}$$

$$\Rightarrow \begin{aligned} -2x + y + 2z &= 0 \\ -4y &= 0 \\ 2x + y - 2z &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} -2x + y + 2z &= 0 \\ 4y &= 0 \\ 2x + y - 2z &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} 2x - y - 2z &= 0 \\ y &= 0 \\ 2x + y - 2z &= 0 \end{aligned}$$ [Multiplying by -1 on both sides]

We get, $y = 0$

$$\Rightarrow \begin{aligned} 2x - y - 2z &= 0 \\ 2x + y - 2z &= 0 \end{aligned}$$

We have,

$L_i \to -a_{i1}L_1 + a_{11}L_i$

$i = 2,\ L_2 \to -a_{21}L_1 + a_{11}L_2$

$$\begin{aligned} &= -2(2x - y - 2z) + 2(2x + y - 2z) \\ &= -4x + 2y + 4z + 4x + 2y - 4z \\ &= 4y \end{aligned}$$

Thus we obtain following system

$\Rightarrow \quad 2x - y - 2z = 0$

$$\mathbf{4y = 0}$$

$$\Rightarrow \quad \mathbf{2x - y - 2z = 0}$$
$$\mathbf{y = 0}$$

$$\Rightarrow \quad \mathbf{2x - 0 - 2z = 0}$$

$$\Rightarrow \quad \mathbf{2x - 2z = 0}$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is z.

Let,

$z = 1$

$\therefore y = 0$ [obtained]

$\therefore 2x - 2 = 0$

$\Rightarrow x = 1$

$$\therefore \mathbf{v_3} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{bmatrix} = \begin{bmatrix} \mathbf{1} \\ \mathbf{0} \\ \mathbf{1} \end{bmatrix} [\text{For } \lambda = \mathbf{7}]$$

Let P be a matrix whose columns are the above Eigen vectors.

$$\mathbf{P} = [\mathbf{v_1} \quad \mathbf{v_2} \quad \mathbf{v_3}]$$

$$\mathbf{P} = \begin{bmatrix} -2 & -1 & 1 \\ 2 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$|\mathbf{P}| = (-2)(0-0) - (-1)(2-0) + 1(2-0)$$
$$|\mathbf{P}| = 2 + 2 = 4$$

Then, $\mathbf{D = P^{-1}AP}$ is the diagonalized matrix whose leading diagonal are the respective Eigen values.

Now

$$\mathbf{P} = \begin{bmatrix} -2 & -1 & 1 \\ 2 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore \text{ Co-factor of } \mathbf{a_{11}(-2) = c_{11}} = +\begin{vmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{1} & \mathbf{1} \end{vmatrix} = \mathbf{0}$$

$$\therefore \text{ Co-factor of } \mathbf{a_{12}(-1) = c_{12}} = -\begin{vmatrix} \mathbf{2} & \mathbf{0} \\ \mathbf{1} & \mathbf{1} \end{vmatrix} = \mathbf{-2}$$

$\therefore$ Co-factor of $\mathbf{a_{13}(1) = c_{13}} = +\begin{vmatrix} \mathbf{2} & \mathbf{0} \\ \mathbf{1} & \mathbf{1} \end{vmatrix} = \mathbf{2}$

$\therefore$ Co-factor of $\mathbf{a_{21}(2) = c_{21}} = -\begin{vmatrix} \mathbf{-1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} \end{vmatrix} = \mathbf{2}$

$\therefore$ Co-factor of $\mathbf{a_{22}(0) = c_{22}} = +\begin{vmatrix} \mathbf{-2} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} \end{vmatrix} = \mathbf{-3}$

$\therefore$ Co-factor of $\mathbf{a_{23}(0) = c_{23}} = -\begin{vmatrix} \mathbf{-2} & \mathbf{-1} \\ \mathbf{1} & \mathbf{1} \end{vmatrix} = \mathbf{1}$

$\therefore$ Co-factor of $\mathbf{a_{31}(1) = c_{31}} = +\begin{vmatrix} \mathbf{-1} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} \end{vmatrix} = \mathbf{0}$

$\therefore$ Co-factor of $\mathbf{a_{32}(1) = c_{32}} = -\begin{vmatrix} \mathbf{-2} & \mathbf{1} \\ \mathbf{2} & \mathbf{0} \end{vmatrix} = \mathbf{2}$

$\therefore$ Co-factor of $\mathbf{a_{33}(1) = c_{33}} = +\begin{vmatrix} \mathbf{-2} & \mathbf{-1} \\ \mathbf{2} & \mathbf{0} \end{vmatrix} = \mathbf{2}$

Co-factor Matrix: $\mathbf{P^C} = \begin{bmatrix} \mathbf{0} & \mathbf{-2} & \mathbf{2} \\ \mathbf{2} & \mathbf{-3} & \mathbf{1} \\ \mathbf{0} & \mathbf{2} & \mathbf{2} \end{bmatrix}$

Adjoint of $\mathrm{P} = (\mathrm{P^C})^{\mathrm{T}} = \begin{bmatrix} 0 & 2 & 0 \\ -2 & -3 & 2 \\ 2 & 1 & 2 \end{bmatrix}$

$$\therefore \mathbf{P^{-1}} = \frac{\mathbf{Adjoint\ of\ P}}{|\mathbf{P}|} = \frac{\mathbf{1}}{\mathbf{4}}\begin{bmatrix} \mathbf{0} & \mathbf{2} & \mathbf{0} \\ \mathbf{-2} & \mathbf{-3} & \mathbf{2} \\ \mathbf{2} & \mathbf{1} & \mathbf{2} \end{bmatrix}$$

$$\boldsymbol{D = P^{-1}AP} = \frac{\mathbf{1}}{\mathbf{4}}\begin{bmatrix} \mathbf{0} & \mathbf{2} & \mathbf{0} \\ \mathbf{-2} & \mathbf{-3} & \mathbf{2} \\ \mathbf{2} & \mathbf{1} & \mathbf{2} \end{bmatrix}\begin{bmatrix} \mathbf{5} & \mathbf{1} & \mathbf{2} \\ \mathbf{0} & \mathbf{3} & \mathbf{0} \\ \mathbf{2} & \mathbf{1} & \mathbf{5} \end{bmatrix}\begin{bmatrix} \mathbf{-2} & \mathbf{-1} & \mathbf{1} \\ \mathbf{2} & \mathbf{0} & \mathbf{0} \\ \mathbf{1} & \mathbf{1} & \mathbf{1} \end{bmatrix}$$

$$= \frac{\mathbf{1}}{\mathbf{4}}\begin{bmatrix} \mathbf{0} & \mathbf{2} & \mathbf{0} \\ \mathbf{-2} & \mathbf{-3} & \mathbf{2} \\ \mathbf{2} & \mathbf{1} & \mathbf{2} \end{bmatrix}\begin{bmatrix} \mathbf{5(-2)+1\times2+2\times1} & \mathbf{5(-1)+1\times0+2\times1} & \mathbf{5\times1+1\times0+2\times1} \\ \mathbf{0(-2)+3\times2+0\times1} & \mathbf{0(-1)+3\times0+0\times1} & \mathbf{0\times1+3\times0+0\times1} \\ \mathbf{2(-2)+1\times2+5\times1} & \mathbf{2(-1)+1\times0+5\times1} & \mathbf{2\times1+1\times0+5\times1} \end{bmatrix}$$

$$= \frac{\mathbf{1}}{\mathbf{4}}\begin{bmatrix} \mathbf{0} & \mathbf{2} & \mathbf{0} \\ \mathbf{-2} & \mathbf{-3} & \mathbf{2} \\ \mathbf{2} & \mathbf{1} & \mathbf{2} \end{bmatrix}\begin{bmatrix} \mathbf{-10+2+2} & \mathbf{-5+0+2} & \mathbf{5+0+2} \\ \mathbf{0+6+0} & \mathbf{0+0+0} & \mathbf{0+0+0} \\ \mathbf{-4+2+5} & \mathbf{-2+0+5} & \mathbf{2+0+5} \end{bmatrix}$$

$$= \frac{\mathbf{1}}{\mathbf{4}}\begin{bmatrix} \mathbf{0} & \mathbf{2} & \mathbf{0} \\ \mathbf{-2} & \mathbf{-3} & \mathbf{2} \\ \mathbf{2} & \mathbf{1} & \mathbf{2} \end{bmatrix}\begin{bmatrix} \mathbf{-6} & \mathbf{-3} & \mathbf{7} \\ \mathbf{6} & \mathbf{0} & \mathbf{0} \\ \mathbf{3} & \mathbf{3} & \mathbf{7} \end{bmatrix}$$

$$= \frac{1}{4}\begin{bmatrix} 0(-6)+2\times 6+0\times 3 & 0(-3)+2\times 0+0\times 3 & 0\times 7+2\times 0+0\times 7 \\ (-2)(-6)+(-3)(6)+2\times 3 & (-2)(-3)+(-3)\times 0+2\times 3 & (-2)\times 7+(-3)\times 0+2\times 7 \\ 2(-6)+1\times 6+2\times 3 & 2(-3)+1\times 0+2\times 3 & 2\times 7+1\times 0+2\times 7 \end{bmatrix}$$

$$= \frac{1}{4}\begin{bmatrix} 0+12+0 & 0+0+0 & 0+0+0 \\ 12-18+6 & 6+6 & -14-0+14 \\ -12+6+6 & -6+0+6 & 14+0+14 \end{bmatrix}$$

$$= \frac{1}{4}\begin{bmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 28 \end{bmatrix}$$

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

$$\therefore \mathbf{D} = \mathbf{P}^{-1}\mathbf{AP} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 7 \end{bmatrix} \textit{ Answer}$$

Home Task:

Diagonalize the matrix

$$i. \begin{bmatrix} 2 & -1 \\ -3 & 4 \end{bmatrix} \quad ii. \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{bmatrix} \quad iii. \begin{bmatrix} 5 & 0 & 0 \\ 1 & 0 & 3 \\ 0 & 0 & -2 \end{bmatrix} \quad iv. \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \quad v. \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad vi. \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$$

Solve the diagonalization problem for $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Hints:

$\lambda = 1,1$

We cannot obtain two linearly independent eigenvectors for the given matrix. Thus the matrix is *not* diagonalizable.

Example 52: Use Elementary row operations to transform the row-reduced matrix to the identity matrix:

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \textbf{(By adding to the second row –3 times the third row)}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \textbf{(By adding to the first row –1 times the third row)}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \textbf{(By adding to the first row –2 times the second row)}$$

Thus, a square matrix A has an inverse if and only if A can be transformed into an identity matrix with elementary row operations.

Finding Inverse making Linear Equations:

Example 53: **Find** A^{-1} **where** $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$

Answer:

Given, $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$, **Let** $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ **and** $B = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$

$\therefore AX = B$

$$\begin{bmatrix} 1 \times x_1 + 2 \times x_2 \\ 3 \times x_1 + 1 \times x_2 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

The matrix can be written as

$x_1 + 2x_2 = d_1 \text{ (say)}$ -----------------(i)

$3x_1 + x_2 = d_2 \text{ (say)}$ -----------------(ii)

$(ii) \times 2 - (i)$:

$5x_1 = 2d_2 - d_1$

$x_1 = -\frac{1}{5}d_1 + \frac{2}{5}d_2$ ----------------(iii)

$(i) \times 3 - (ii)$:

$5x_2 = 3d_1 - d_2$

$x_2 = \frac{3}{5}d_1 - \frac{1}{5}d_2$--------------------(iv)

FROM (iii) and (iv) we get,

$$\therefore A^{-1} = \begin{bmatrix} -\frac{1}{5} & \frac{2}{5} \\ \frac{3}{5} & -\frac{1}{5} \end{bmatrix}$$

Question: Define Augmented Matrix with example:

Answer: **The augmented matrix for AX=b is the partitioned matrix** $[A|b]$**.**

All the scalars contained in the system of equations AX=b appear in the coefficient matrix A and the column matrix b. These scalars can be combined into the single partitioned matrix $[A|b]$, known as the augmented matrix for the system of equations.

Example 54: $\mathbf{x_1 + x_2 - 2x_3 = -3}$

$$\mathbf{2x_1 + 5x_2 + 3x_3 = 11}$$

$$\mathbf{-x_1 + 3x_2 + x_3 = 5}$$

can be written as the matrix equation

Let, $\mathbf{A} = \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{-2} \\ \mathbf{2} & \mathbf{5} & \mathbf{3} \\ \mathbf{-1} & \mathbf{3} & \mathbf{1} \end{bmatrix}$; $\quad \mathbf{X} = \begin{bmatrix} \mathbf{x_1} \\ \mathbf{x_2} \\ \mathbf{x_3} \end{bmatrix}$; $\quad \mathbf{b} = \begin{bmatrix} \mathbf{-3} \\ \mathbf{11} \\ \mathbf{5} \end{bmatrix}$

AX= b

$$\begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{-2} \\ \mathbf{2} & \mathbf{5} & \mathbf{3} \\ \mathbf{-1} & \mathbf{3} & \mathbf{1} \end{bmatrix} \begin{bmatrix} \mathbf{x_1} \\ \mathbf{x_2} \\ \mathbf{x_3} \end{bmatrix} = \begin{bmatrix} \mathbf{-3} \\ \mathbf{11} \\ \mathbf{5} \end{bmatrix}$$

Which has as its augmented matrix?

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 1 & -2 & -3 \\ 2 & 5 & 3 & 11 \\ -1 & 3 & 1 & 5 \end{array}\right]$$

Elementary Row and Column Operations

The matrix operations of

1. Interchanging two rows or columns,
2. Adding a multiple of one row or column to another,
3. Multiplying any row or column by a nonzero element.

Calculating Inverses by the elementary row operation method

Step1: Create an augment matrix $[A|I]$, where A is the n×n matrix to be inverted and I is the n×n identity matrix.

Step 2: Use elementary row operations on $[A|I]$ to transform the left partition A to row-reduced form, applying each operation to the full augmented matrix.

Step 3: If the left partition of the row-reduced matrix has zero elements on its main diagonal, stop-A does not have an inverse-otherwise, continue.

Step 4: Use elementary row operations on the row-reduced augmented matrix to transform the left partition to the n×n identity matrix, applying each operation to the full augmented matrix.

Step 5: The right partition of the final augmented matrix is the inverse of A.

[Note: It is easy to show that if any diagonal element in a diagonal matrix is (zero) 0, then that matrix is singular.]

Example 55: Find the inverse of $\mathbf{A}=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$

Solution: $[A|I]=\left[\begin{array}{cc|cc}1 & 2 & 1 & 0\\ 3 & 4 & 0 & 1\end{array}\right]$

$r_2{}' = r_2 + (-3)r_1$

$$\approx\left[\begin{array}{cc|cc}1 & 2 & 1 & 0\\ 3+(-3).1 & 4+(-3).2 & 0+(-3).1 & 1+(-3).0\end{array}\right]$$

$$\approx\left[\begin{array}{cc|cc}1 & 2 & 1 & 0\\ 3-3 & 4-6 & -3 & 1+0\end{array}\right]$$

$$\approx\left[\begin{array}{cc|cc}1 & 2 & 1 & 0\\ 0 & -2 & -3 & 1\end{array}\right]$$

$r_2{}' = r_2 \times (-\frac{1}{2})$

$$\approx\left[\begin{array}{cc|cc}1 & 2 & 1 & 0\\ 0\times(-\frac{1}{2}) & -2\times(-\frac{1}{2}) & -3\times(-\frac{1}{2}) & 1\times(-\frac{1}{2})\end{array}\right]$$

$$\approx\left[\begin{array}{cc|cc}1 & 2 & 1 & 0\\ 0 & 1 & \frac{3}{2} & -\frac{1}{2}\end{array}\right]$$

$r_1{}' = r_1 + (-2)r_2$

$$\approx\left[\begin{array}{cc|cc}1+(-2).0 & 2+(-2).1 & 1+(-2)\frac{3}{2} & 0+(-2)(-\frac{1}{2})\\ 0 & 1 & \frac{3}{2} & -\frac{1}{2}\end{array}\right]$$

$$\approx\left[\begin{array}{cc|cc}1+0 & 2-2 & 1-3 & 0+1\\ 0 & 1 & \frac{3}{2} & -\frac{1}{2}\end{array}\right]$$

$$\approx\left[\begin{array}{cc|cc}1 & 0 & -2 & 1\\ 0 & 1 & \frac{3}{2} & -\frac{1}{2}\end{array}\right]$$

Thus $A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & \frac{-1}{2} \end{bmatrix}$

Example 56: **Find the inverse of** $A = \begin{bmatrix} 5 & 8 & 1 \\ 0 & 2 & 1 \\ 4 & 3 & -1 \end{bmatrix}$

Solution: $\left[\begin{array}{ccc|ccc} 5 & 8 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 4 & 3 & -1 & 0 & 0 & 1 \end{array}\right]$

$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0.2 & 0.2 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 4 & 3 & -1 & 0 & 0 & 1 \end{array}\right]$ **(By multiplying the first row by 0.2)**

$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0.2 & 0.2 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & -3.4 & -1.8 & -0.8 & 0 & 1 \end{array}\right]$ **(By adding to the third row –4 times the first row)**

$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0.2 & 0.2 & 0 & 0 \\ 0 & 1 & 0.5 & 0 & 0.5 & 0 \\ 0 & -3.4 & -1.8 & -0.8 & 0 & 1 \end{array}\right]$ **(By multiplying the second row by** $\frac{1}{2}$**)**

$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0.2 & 0.2 & 0 & 0 \\ 0 & 1 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & -0.1 & -0.8 & 1.7 & 1 \end{array}\right]$ **(By adding to the third row 3.4 times the second row)**

$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0.2 & 0.2 & 0 & 0 \\ 0 & 1 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 1 & 8 & -17 & -10 \end{array}\right]$ **(by multiplying the third row by –10)**

A has been transformed into row-reduced form with a main diagonal of only ones; A has an inverse. Continuing with the transformation process, we get:

$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0.2 & 0.2 & 0 & 0 \\ 0 & 1 & 0 & -4 & 9 & 5 \\ 0 & 0 & 1 & 8 & -17 & -10 \end{array}\right]$ (By adding to the second row –0.5 times the third row)

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.6 & 0 & -1.4 & 3.4 & 2 \\ 0 & 1 & 0 & -4 & 9 & 5 \\ 0 & 0 & 1 & 8 & -17 & -10 \end{array}\right]$$ (by adding to the first row –0.2 times the third row)

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & -11 & -6 \\ 0 & 1 & 0 & -4 & 9 & 5 \\ 0 & 0 & 1 & 8 & -17 & -10 \end{array}\right]$$ (by adding to the first row –1.6 times the second row)

Thus, $\mathbf{A}^{-1} = \begin{bmatrix} 5 & -11 & -6 \\ -4 & 9 & 5 \\ 8 & -17 & -10 \end{bmatrix}$

Question: State Cayely Hamilton Theorem with example:

Answer:

Statement: Every Square matrix satisfies its characteristic equation or if $|\mathbf{A} - \lambda \mathbf{I}| = (-1)^n [\lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + ---------- a_n]$ be the characteristic polynomial of n×n matrix $A = [a_{ij}]$, then the matrix equation $\mathbf{X}^n + a_1 \mathbf{X}^{n-1} + --------- a_n \mathbf{I} = 0$ is satisfied by X=A

i.e $\mathbf{A}^n + a_1 \mathbf{A}^{n-1} + -------------- a_n \mathbf{I} = 0$

Example 57: Show that the matrix A=$\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ satisfies Caley-Hamilton Theorem and hence compute $\mathbf{A}^{-1}$

Answer: Hence we have,

$$\mathbf{A} - \lambda \mathbf{I} = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 4-0 \\ 2-0 & 3-\lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix}$$

$|A-\lambda I| = (1-\lambda)(3-\lambda) - 8$

$|A-\lambda I| = (3-\lambda-3\lambda+\lambda^2) - 8$

$|A-\lambda I| = 3-4\lambda+\lambda^2 - 8$

$|A-\lambda I| = \lambda^2 - 4\lambda - 5$

$\therefore$ **Characteristic equation of A is** $|A-\lambda I| = 0$

$$\lambda^2 - 4\lambda - 5 = 0 \text{ ------------------------------(i)}$$

Replace λ by A in L.H.S of (i)

Now $A^2 - 4A - 5I$

$$= \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 5\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+8 & 4+12 \\ 2+6 & 8+9 \end{bmatrix} - \begin{bmatrix} 4 & 16 \\ 8 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix} - \begin{bmatrix} 4 & 16 \\ 8 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0 = \textbf{NULL MATRIX}$$

That is $A^2 - 4A - 5I = 0$ ------------------------------(ii)

Hence the given matrix A satisfies its characteristic equation (i) i.e satisfy Caley Hamilton Theorem.

Again, from (ii)

$A^2 - 4A - 5I = 0$ **gives**

$A^2 - 4A = 5I$ ------------------------------(iii)

Multiplying (iii) by A^{-1} on both sides:

$A^2.A^{-1} - 4AA^{-1} = 5A^{-1}I$

$A - 4I = 5A^{-1}$

$[\because AA^{-1} = I \text{ \& } A^{-1}I = A^{-1}]$

Example:

$$\text{Let, } \mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \ \mathbf{X} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\therefore \mathbf{IX} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1\times x + 0\times y + 0\times z \\ 0\times x + 1\times y + 0\times z \\ 0\times x + 0\times y + 1\times z \end{pmatrix}$$

$$= \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$= \mathbf{X}$$

$$\therefore \mathbf{IX} = \mathbf{X}$$

$\mathbf{A - 4I = 5A^{-1}}$

$\therefore \mathbf{A^{-1} = \frac{1}{5}(A - 4I)}$ ------------------------------(iv)

Putting the value of A and I in (iv), we get

$$\therefore \mathbf{A^{-1} = \frac{1}{5}(A - 4I)}$$

$$\therefore \mathbf{A^{-1}} = \frac{1}{5}\left\{\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\}$$

$$\therefore A^{-1} = \frac{1}{5}\left\{\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}\right.$$

$$\therefore A^{-1} = \frac{1}{5}\left\{\begin{bmatrix} 1-4 & 4-0 \\ 2-0 & 3-4 \end{bmatrix}\right.$$

$$= \frac{1}{5}\begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

<u>Perfection Test:</u>

$$AA^{-1} = \frac{1}{5}\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

$$AA^{-1} = \frac{1}{5}\begin{bmatrix} 1\times(-3)+4\times 2 & 1\times 4+4\times(-1) \\ 2\times(-3)+3\times 2 & 2\times 4+3\times(-1) \end{bmatrix}$$

$$AA^{-1} = \frac{1}{5}\begin{bmatrix} -3+8 & 4-4 \\ -6+6 & 8-3 \end{bmatrix}$$

$$AA^{-1} = \frac{1}{5}\begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$AA^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AA^{-1} = I$$

Home Task

Show that the matrix $\begin{bmatrix} 2 & 0 & 1 \\ -2 & 3 & 4 \\ -5 & 5 & 6 \end{bmatrix}$ **satisfies Caley-Hamilton Theorem and hence compute** $\mathbf{A^{-1}}$

Geometrical Applications of Matrices

A matrix is a rectangular array of numbers, which can be used to give information. The coordinates in two dimensions of a point can be represented by a matrix, so that the coordinates (2,3) of a point can be represented by a matrix (2,3) or by a matrix $\begin{pmatrix} \mathbf{2} \\ \mathbf{3} \end{pmatrix}$.

The components of a vector can be represented by a matrix
Example:

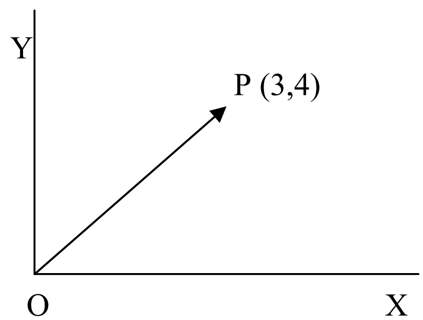


Figure no 26

$$\vec{\mathbf{OP}} = \begin{pmatrix} \mathbf{3} \\ \mathbf{4} \end{pmatrix} = (\mathbf{3}\,\hat{\mathbf{i}} + \mathbf{4}\,\hat{\mathbf{j}})$$

We can even write the coordinates of several points in a single matrix, so that the points coordinates O(0,0), P(1,0), Q(1,1) and R(0,1), the vertices of the unit square, can be displayed in the matrix $\begin{pmatrix} \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} \end{pmatrix}$(Fig:1) and the points O(0,0), P(2,0), Q(2,1) and R(0,1) can be displayed in the matrix $\begin{pmatrix} \mathbf{0} & \mathbf{2} & \mathbf{2} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} \end{pmatrix}$ (Fig:2). Since these points are the vertices of a rectangle, call this matrix R.

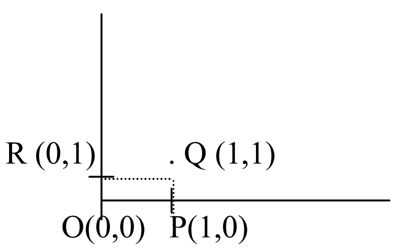


Figure no 27

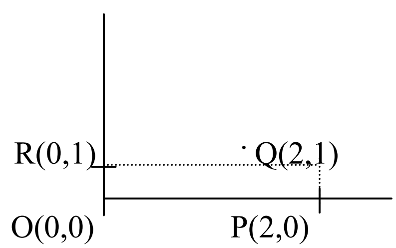


Figure no 28

When the matrix R is multiplied by 2× 2-unit matrixes, it is unaltered.

$$\begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{pmatrix}\begin{pmatrix} \mathbf{0} & \mathbf{2} & \mathbf{2} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{2} & \mathbf{2} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} \end{pmatrix}$$

Reflection:

When R is multiplied by $\begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{-1} \end{pmatrix}$,

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 2 & 0 \\ 0 & 0 & -1 & -1 \end{pmatrix}.$$

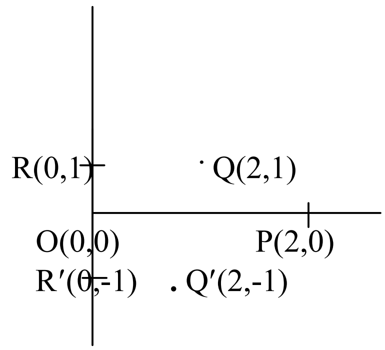


Figure no 29

The rectangle OPQR has been reflected in the x-axis. The rectangle OPQ′R′ is called the image of OPQR under the transformation described by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ or more simply, the image under the transformation $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Rotation:

Similarly $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & -2 & -2 & 0 \end{pmatrix}$ showing that OPQR has been rotated about the origin through 90^0 in a clockwise sense, and we see that the matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ describes a rotation through 90^0 in a clockwise sense. The matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ describes a rotation about the origin through 90^0 in an anticlockwise sense.

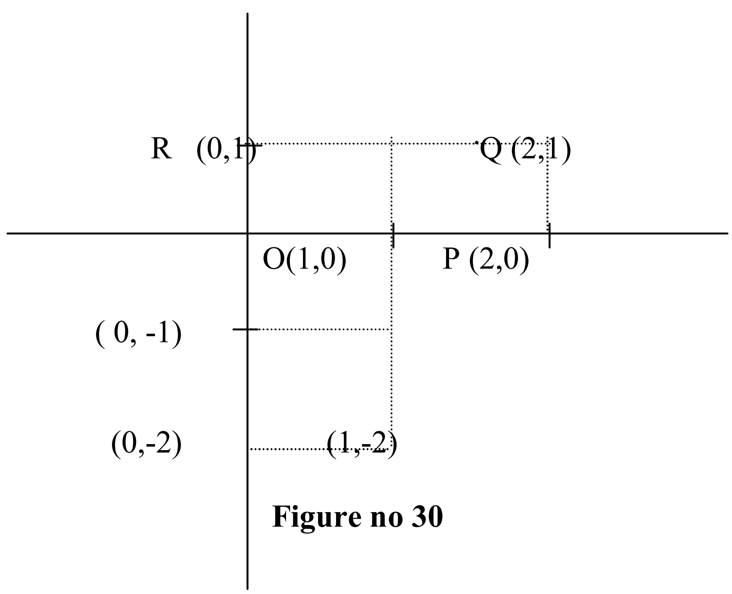


Figure no 30

Enlargement: If we consider the effect of the matrix $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$ on the unit square,

$$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 3 & 3 & 0 \\ 0 & 0 & 3 & 3 \end{pmatrix}$$

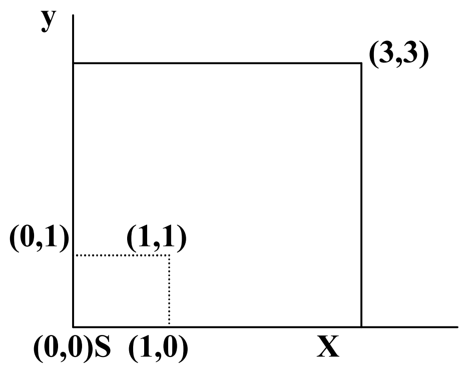


Figure no 31

We see that each side has been enlarged by a factor 3 and the area has been increased by 9, the figure remaining a square. Thus we see that matrices can describe some of the simplest geometrical operations, reflection, rotation and enlargement. We should, of course, show that these matrices would make the appropriate transformation on any figure, not just on the unit square and rectangle.

Example 58: Given that $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, find $A^2, A^3 \textit{ and } A^4$. Investigate the transformations described by these matrices.

Answer: $\mathbf{A}^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

$$\mathbf{A}^3 = \mathbf{A}.\mathbf{A}^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$\mathbf{A}^4 = \mathbf{A}.\mathbf{A}^3 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

The coordinates of any point P (x,y) can be written as a matrix $\begin{pmatrix} x \\ y \end{pmatrix}$, denoted by X and we see that:

$$\mathbf{A}.\mathbf{X} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

$$\mathbf{A}^2.\mathbf{X} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix}$$

$$\mathbf{A}^3.\mathbf{X} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ -x \end{pmatrix}$$

and $A^4.X = X$, *Since* A^4 is the 2×2 unit matrix.

The matrix A has rotated OP through 90^0 in the anticlockwise sense, A^2 has rotated OP through 180^0, A^3 has rotated OP through 270^0 *and* A^4 has rotated OP through 360^0, mapping OP onto itself.

Another Example:

$$\begin{pmatrix} \mathbf{1} & \mathbf{k} \\ \mathbf{0} & \mathbf{1} \end{pmatrix}\begin{pmatrix} \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{1} & \mathbf{1+k} & \mathbf{k} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{1} \end{pmatrix}$$

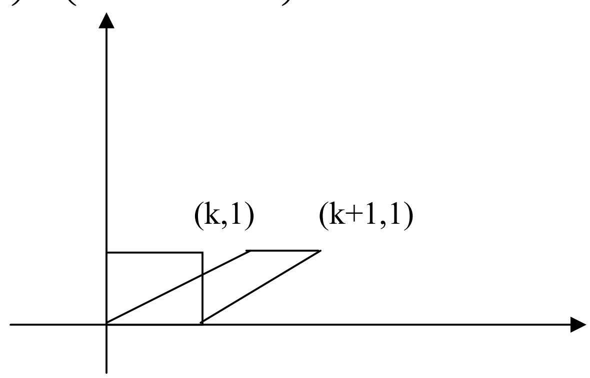


Figure no 32

The matrix $\begin{pmatrix} \mathbf{1} & \mathbf{k} \\ \mathbf{0} & \mathbf{1} \end{pmatrix}$ has mapped the unit square into a parallelogram as the above figure.

Interpret the meaning of a determinant:

There are many ways to interpret the meaning of a determinant. One simple interpretation concerns 2D parallelograms and 3D parallelepipeds. In the 2D case,

Example 59: Consider the vectors $\vec{\mathbf{u}} = (\mathbf{u_1}, \mathbf{u_2})$ and $\vec{\mathbf{v}} = (\mathbf{v_1}, \mathbf{v_2})$. Then the area of the parallelogram
formed by ***u*** and ***v***,

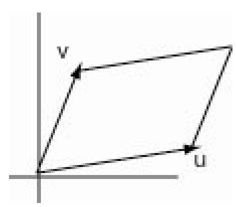


Figure no 33

$$\text{Area } = \begin{vmatrix} \mathbf{u_1} & \mathbf{v_1} \\ \mathbf{u_2} & \mathbf{v_2} \end{vmatrix} = \mathbf{u_1 v_2 - u_2 v_1}$$

Similarly, with three 3D vectors $(\mathbf{u_1}, \mathbf{u_2}, \mathbf{u_3})$ $(\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3})$ and $(\mathbf{w_1}, \mathbf{w_2}, \mathbf{w_3})$ we can define a parallelepiped, whose volume is

$$\text{Volume } = \begin{vmatrix} \mathbf{u_1} & \mathbf{v_1} & \mathbf{w_1} \\ \mathbf{u_2} & \mathbf{v_2} & \mathbf{w_2} \\ \mathbf{u_3} & \mathbf{v_3} & \mathbf{w_3} \end{vmatrix}$$

Determinants have many uses, but we will primarily use them to define and evaluate the cross product of two 3D vectors

Example 60: **Find the area of the parallelogram defined by the column matrices**
$u = \begin{bmatrix} -4 \\ 4 \end{bmatrix}$ and $v = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

Answer: $\left| \mathbf{determinant} \begin{bmatrix} \mathbf{-4} & \mathbf{6} \\ \mathbf{4} & \mathbf{2} \end{bmatrix} \right| = |-\mathbf{32}| = \mathbf{32}$ square units.

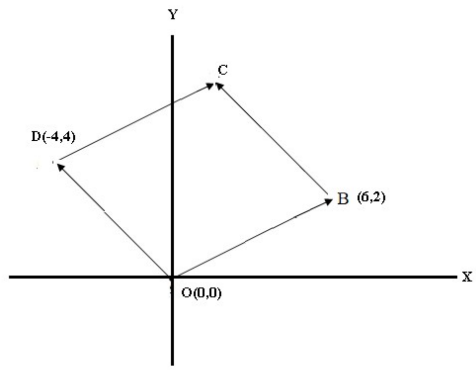


Figure no 34

Example 61: **Find the area of the parallelogram defined by the column matrices**
$\mathbf{u} = \begin{bmatrix} \mathbf{-3} \\ \mathbf{-1} \end{bmatrix}$ **and** $\mathbf{v} = \begin{bmatrix} \mathbf{6} \\ \mathbf{2} \end{bmatrix}$

Answer: $\left| \mathbf{determinant} \begin{bmatrix} \mathbf{-3} & \mathbf{6} \\ \mathbf{-1} & \mathbf{2} \end{bmatrix} \right| = |\mathbf{0}| = \mathbf{0}$ **square units.**

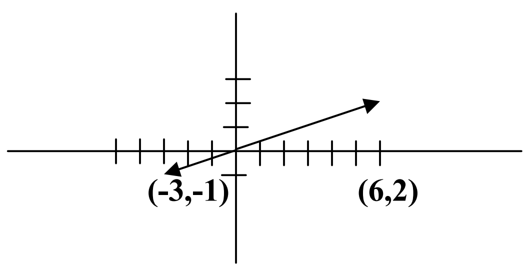


Figure no 35

Some Important Properties:

01) The determinant of an **upper or lower triangular matrix** is the product of the elements on the main diagonal.

Example 62: $\det\begin{bmatrix} 2 & 6 & -4 & 1 \\ 0 & 5 & 7 & -4 \\ 0 & 0 & -5 & 8 \\ 0 & 0 & 0 & 3 \end{bmatrix} = 2(5)(-5)(3) = -150$

02) The determinant of a diagonal matrix is the product of the elements on its main diagonal

03) If a square matrix has a zero row or a zero column, then its determinant is 0.

04) If two rows of a square matrix are identical, then its determinant is 0.

05) A square matrix has an inverse if and only if its determinant is nonzero.

06) The sum of the eigenvalues of a matrix equals the trace of the matrix

Example 63: Determine whether $\lambda_1 = 12$ **and** $\lambda_2 = -4$ are eigen values for $A = \begin{bmatrix} 11 & 3 \\ -5 & -5 \end{bmatrix}$

Solution: Here tr(A)=11+(-5)=6 ≠8 $=\lambda_1 + \lambda_2$, So these numbers are not the eigen values of A.

07) The eigenvalues of an upper or lower triangular matrix are the elements on the main diagonal

Example 64: The matrix $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & -1 \end{bmatrix}$ is lower triangular, so its eigenvalues are $\lambda_1 = \lambda_2 = 1$ **and** $\lambda_3 = -1$

08) A matrix is singular if and only if it has a zero eigenvalue

09) The product of all the eigenvalues of a matrix equals to the determinant of the matrix

Example 65: The eigenvalues of $A = \begin{bmatrix} 11 & 3 \\ -5 & -5 \end{bmatrix}$ are $\lambda_1 = 10$ **and** $\lambda_2 = -4$

Here det(A)=-55+15=-40=$\lambda_1\lambda_2$

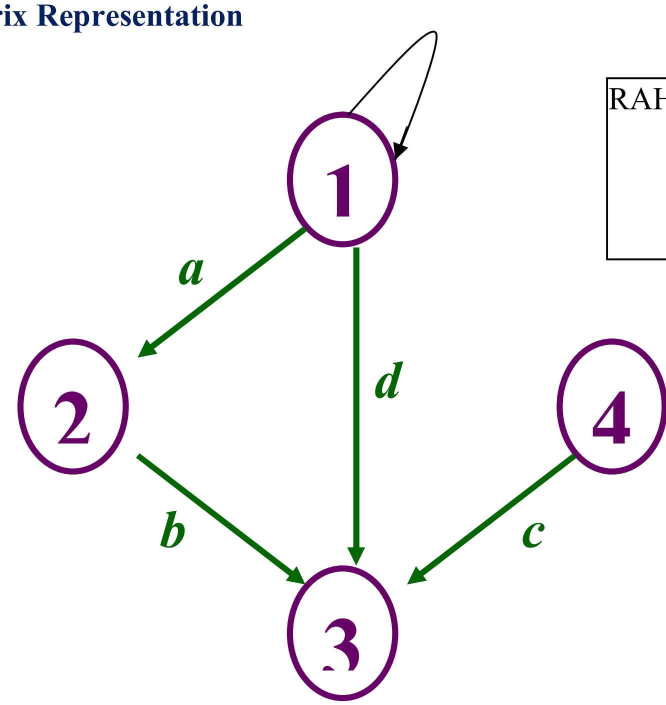
10) If λ is an eigenvalue of A, then λ also an eigenvalue of A^T

11) An n×n matrix is diagonalizable if and only if the matrix possesses n linearly independent eigenvectors

Representing Graphs Using Adjacency Matrix:

Adjacency Matrix Representation

Example 66:



RAHIM

KARIM

Figure no 36

A	1	2	3	4
1	1	1	1	0
2	0	0	1	0
3	0	0	0	0
4	0	0	1	0

An *adjacency matrix* represents the graph as a n x n matrix A:

A [i, j] = 1 if edge (i, j) εE (or weight of edge)

$= 0$ if edge (i, j) εE

Node:

- Nodes contain data and/or links to other nodes. Links between nodes are often implemented by pointers or references.
- Multiple computers are connected with a network in a distributed processing system (or a parallel processing system). Each computer of such a system is called a `node'.
- Any device that is directly connected to the network, usually through Ethernet cable. Nodes include file servers and shared peripherals.
- A node is a device that is connected as part of a computer network. Nodes can be computers, personal digital assistants (PDAs), cell phones, or various other network appliances.
- The place on a stem where the leaves or branches are attached.
- A single machine on a network.
- The point on a plant stem where a leaf or leaves are attached; the point on a stem from which new leaves or stems will grow.
- A device attached to a network. A node uses the network as a means of communication and has an address on the network
- End point of a network connection. Nodes include any device attached to a network such as file servers, printers, or workstations.
- The part of the stem from which one or more leaves arise.
- The point on a stem where a leaf or bud is attached.
- This is any point of connection in a network.
- The point of intersection of a planet's orbit and the ecliptic.
- A linked list is a linear collection of data elements (called nodes), where the linear order is given by means of pointers.
- When two or more devices connected for the purpose of sharing data or resources can form a network. Devices referred to as nodes.
- A linked list is one of the fundamental data structures used in computer programming

Different of linked list: Doubly linked list, singly linked list, linked list, double linked list, circularly linked list, unrolled linked list

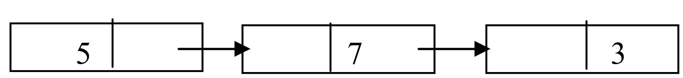
Each node is divided into 2 parts in linked list:

i) 1st part contains the information of the element

ii) 2nd part is called the link field or next pointer field, which contains the address of the next node in the list.

(In link list)

```
Struct node {
        int value;
        struct node *next;
    }
```



Undirected Graph:

A graph $G = (V, E)$

V: vertices

E : edges, unordered pairs of vertices from $V \times V$

(u,v) is same as (v,u)

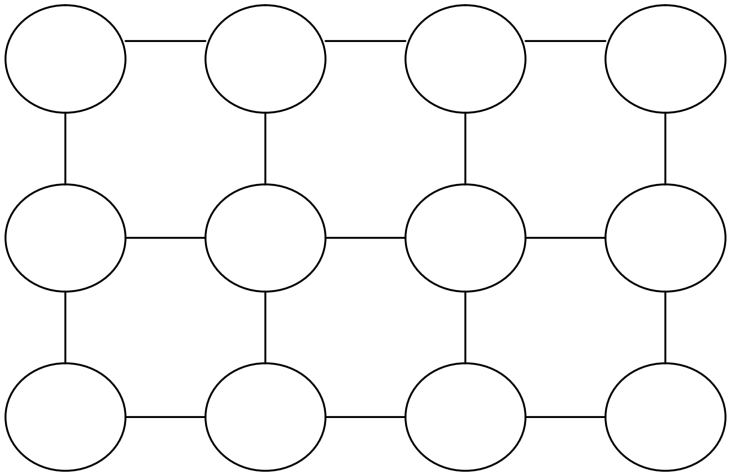


Figure no 37

Directed Graph:

A graph $G = (V, E)$

V: vertices

E: edges, ordered pairs of vertices from $V \rightarrow V$

F(u,v) is different from (v,u)

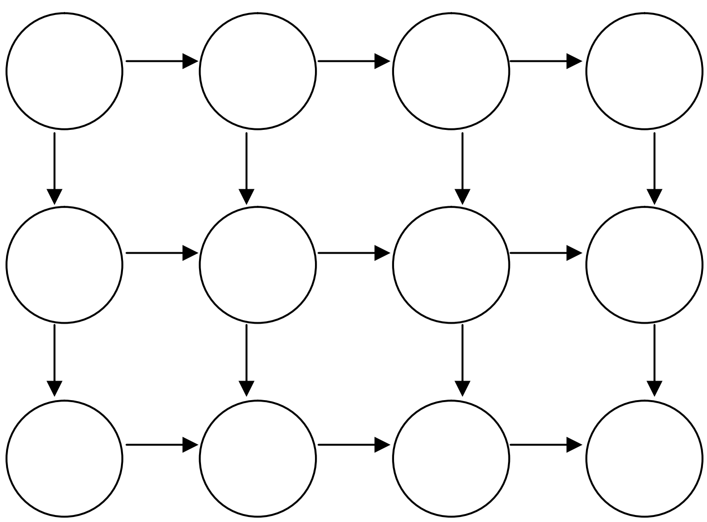


Figure no 38

Strongly Connected Components:

- Every pair of vertices are reachable from each other
- Graph G is ***strongly connected*** if, for every u and v in V, there is some path from u to v and some path from v to u.

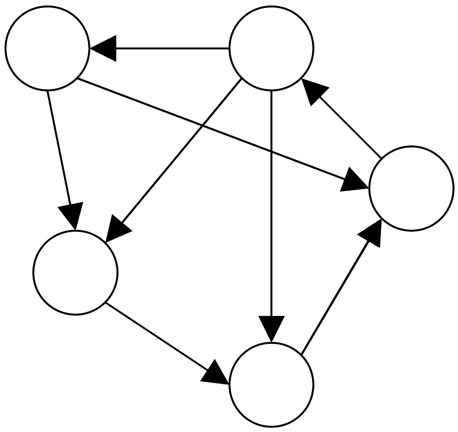


Figure no 39: Strongly Connected

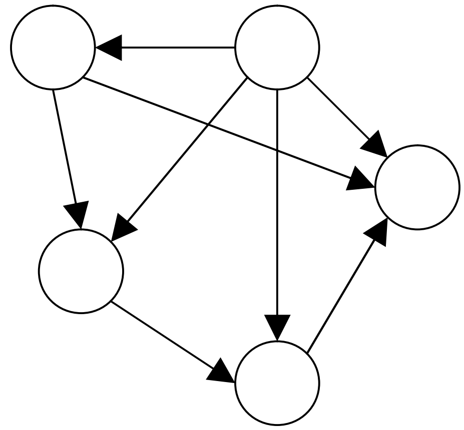
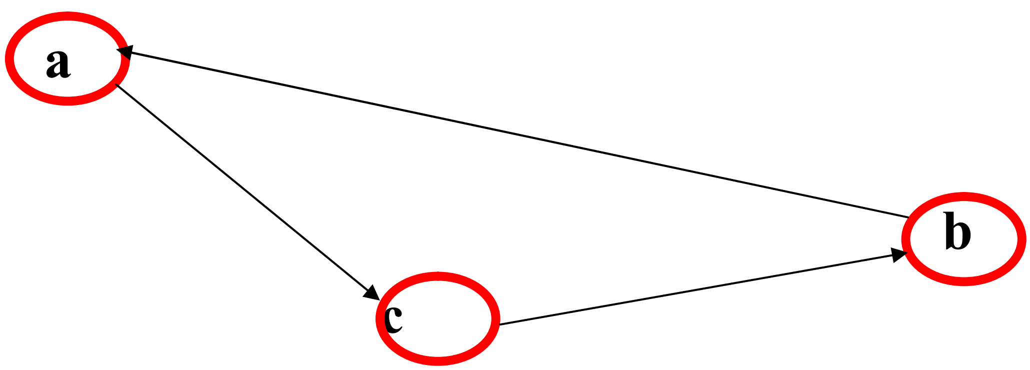


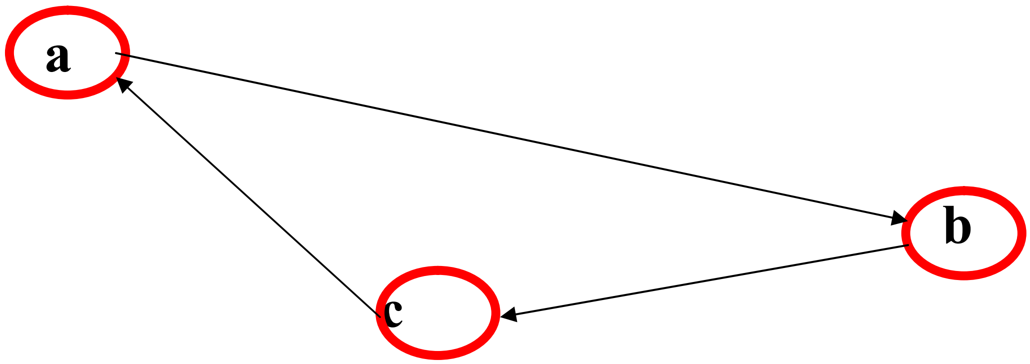
Figure no 40: Not Strongly Connected

Graphically Representation of Transpose of a Matrix:

The graph G^T is the transpose of G, which is visualized by reversing the arrows on the digraph.



Graph (G)
Figure no 42



Graph (G^T)

Figure no 43

	a	b	c
a	0	0	1
b	1	0	0
c	0	1	0

Matrix of Graph G

	a	b	c
a	0	1	0
b	0	0	1
c	1	0	0

Matrix of Graph G^T

Matrix Decomposition Theorem

Matrix **decomposition** is a factorization of a matrix into some canonical form (Standard form/Normal form). There are many different matrix decompositions; each finds use among a particular class of problems.

Decompositions related to solving systems of linear equations

- LU decomposition
- Cholesky decomposition
- QR decomposition

Decompositions based on eigenvalues and related concepts

- **Eigen decomposition (also called spectral/diagonal decomposition)**
- Singular value decomposition

A. Eigen/diagonal Decomposition or spectral decomposition Theorem:

In linear algebra, eigendecomposition or sometimes spectral decomposition is the factorization of a matrix into a canonical form, whereby the matrix is represented in terms of its eigenvalues and eigenvectors. Only diagonalizable matrices can be factorized in this way.

Let A be the given Matrix, D is a diagonal matrix with the corresponding eigen values of Matrix A and **P be a matrix of eigenvectors** of a given Matrix A, Then A can be written as an eigen decomposition $\mathbf{A = PDP^{-1}}$

Example 67: Examine the eigen decomposition for the matrix $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

Let $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

$$\therefore \mathbf{A - \lambda I} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{pmatrix}$$

$$|\mathbf{A - \lambda I}| = \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = (4-\lambda)(4-\lambda) - 1$$

$$\Rightarrow \quad |\mathbf{A - \lambda I}| = 0$$

$$\Rightarrow \quad \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = (4-\lambda)(4-\lambda) - 1 = 0$$

$$\Rightarrow (16 - 4\lambda - 4\lambda + \lambda^2) - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 16 - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 15 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda - 3\lambda + 15 = 0$$

$$\Rightarrow \lambda(\lambda - 5) - 3(\lambda - 5) = 0$$

$$\Rightarrow (\lambda - 5)(\lambda - 3) = 0$$

$$\Rightarrow \lambda = 5, \lambda = 3$$

$$\therefore \mathbf{D} = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$$

Now to find out **the eigenvector** of the matrix A,

Let a non-zero vector $\mathrm{v} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}$

We have, $\mathbf{Av = \lambda v}$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = 3\begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 3]$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \end{pmatrix}$$

$$\Rightarrow \begin{aligned} 4x+y &= 3x \\ x+4y &= 3y \end{aligned}$$

$$\Rightarrow \begin{aligned} x+y &= 0 \\ x+y &= 0 \end{aligned}$$

$$\Rightarrow x + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Let y = 1 then x = -1

$\therefore$ Eigenvector $\mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for $\lambda = 3$

Again for $\lambda = 5$

Let a non-zero vector v = $\begin{bmatrix} x \\ y \end{bmatrix}$

We have, $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 5\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = 5\begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 5]$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = \begin{pmatrix} 5x \\ 5y \end{pmatrix}$$

$$\Rightarrow \begin{aligned} 4x+y &= 5x \\ x+4y &= 5y \end{aligned}$$

$$\Rightarrow \begin{aligned} -x+y &= 0 \\ x-y &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} x-y &= 0 \\ x-y &= 0 \end{aligned}$$

$$\Rightarrow x - y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular **(2-1) =1 free variable**, which is y.

Let y =1 , then x =1

$\therefore$ Eigenvector $\mathbf{v} = \begin{pmatrix} \mathbf{1} \\ \mathbf{1} \end{pmatrix}$ for $\lambda = \mathbf{5}$

P be a matrix of eigenvectors of a given Matrix A.

$$\therefore \mathbf{P} = \begin{pmatrix} -\mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} \end{pmatrix}$$

Now to find out $\mathbf{p}^{-1}$

$$\mathbf{c_{11} = 1,\ c_{12} = -1,\ c_{21} = -1,\ c_{22} = -1}$$

$$\mathbf{p^c} = \begin{pmatrix} \mathbf{1} & -\mathbf{1} \\ -\mathbf{1} & -\mathbf{1} \end{pmatrix}$$

$$\mathbf{Adj\,P = (P^C)^T} = \begin{pmatrix} \mathbf{1} & -\mathbf{1} \\ -\mathbf{1} & -\mathbf{1} \end{pmatrix}$$

$$|\mathbf{P}| = \begin{vmatrix} -\mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} \end{vmatrix} = \mathbf{-1 - 1 = -2}$$

$$\therefore \mathbf{P^{-1}} = \frac{\mathbf{AdjP}}{|\mathbf{P}|} = -\frac{\mathbf{1}}{\mathbf{2}}\begin{pmatrix} \mathbf{1} & -\mathbf{1} \\ -\mathbf{1} & -\mathbf{1} \end{pmatrix}$$

Then A can be written as an eigen decomposition $\mathbf{A = PDP^{-1}}$

<u>Justification:</u>

$$\begin{pmatrix} \mathbf{4} & \mathbf{1} \\ \mathbf{1} & \mathbf{4} \end{pmatrix} = -\frac{\mathbf{1}}{\mathbf{2}}\begin{pmatrix} -\mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} \end{pmatrix}\begin{pmatrix} \mathbf{3} & \mathbf{0} \\ \mathbf{0} & \mathbf{5} \end{pmatrix}\begin{pmatrix} \mathbf{1} & -\mathbf{1} \\ -\mathbf{1} & -\mathbf{1} \end{pmatrix}$$

$$= -\frac{\mathbf{1}}{\mathbf{2}}\begin{pmatrix} -\mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} \end{pmatrix}\begin{pmatrix} \mathbf{3} & -\mathbf{3} \\ -\mathbf{5} & -\mathbf{5} \end{pmatrix}$$

$$= -\frac{\mathbf{1}}{\mathbf{2}}\begin{pmatrix} \mathbf{-3-5} & \mathbf{3-5} \\ \mathbf{3-5} & \mathbf{-3-5} \end{pmatrix}$$

$$= -\frac{\mathbf{1}}{\mathbf{2}}\begin{pmatrix} -\mathbf{8} & -\mathbf{2} \\ -\mathbf{2} & -\mathbf{8} \end{pmatrix}$$

$$= \begin{pmatrix} \mathbf{4} & \mathbf{1} \\ \mathbf{1} & \mathbf{4} \end{pmatrix} = \mathbf{A} \quad \text{(Proved)}$$

$\mathbf{A = PDP^{-1}}$ (Proved)

Home task: Find eigen decomposition of the matrix $= \begin{pmatrix} 7 & 3 \\ 3 & -1 \end{pmatrix}$ and $= \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$

Q-01: What is the difference between Eigen decomposition and SVD?

The SVD always exists for any sort of rectangular or square matrix, whereas the eigen decomposition can only exists for square matrices, and even among square matrices sometimes it doesn't exist.

What is singular value of a Matrix?

The singular values of a $\mathbf{m \times n}$ matrix A are the square roots of the eigenvalues of the symmetric $\mathbf{n \times n}$ matrix $\mathbf{A^T A}$. The Eigen values of $\mathbf{A^T A}$ are
$\lambda_1 \geq \lambda_2 \ldots\ldots\ldots\ldots \geq \lambda_r > \lambda_{r+1} \ldots\ldots\ldots\ldots \geq \lambda_n$.

Since large eigenvalues carry significant representation of the matrix, we arrange the values of the Eigen Values from the largest to the smallest and drop/ignore the values towards the last since small eigenvalues are less indicator.

$\sigma_i = \sqrt{\lambda_i}$ are called the singular value of A where i = 1, 2,……………, n.

It is customary to denote the singular values by $\sigma_1, \sigma_2, \sigma_3, \ldots\ldots\ldots\sigma_n$ and to list them in decreasing order:

$\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \ldots\ldots\ldots \geq \sigma_n$

Singular Value Decomposition:

In linear algebra, the **singular value decomposition** (**SVD**) is an important factorization of a rectangular real or complex matrix, with several applications **in signal processing , image processing and compression, gene expression analysis and statistics**. Applications which employ the SVD include computing the pseudo-inverse, matrix approximation, and determining the rank, range and null space of a matrix.

The spectral theorem says that normal matrices, which by definition must be square, can be unitarily diagonalized using a basis of eigenvectors. The SVD can be seen as a generalization of the spectral theorem to arbitrary, not necessarily square, matrices.

The singular value decomposition (SVD) is a generalization of the eigen-decomposition, which can be used to analyze rectangular matrices (the eigen- decomposition is defined only for squared matrices). By analogy with the eigen-decomposition, which decomposes a matrix into two simple matrices, the main idea of the SVD is to decompose a rectangular matrix into three simple matrices: Two orthogonal matrices and one diagonal matrix.

Matrix A is decomposed using SVD to acquire an orthogonal base in the multidimensional feature space:

$$\underbrace{\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} * & \\ & * \\ & \end{bmatrix}}_{S} \underbrace{\begin{bmatrix} * & * \\ * & * \end{bmatrix}}_{V^T}$$

$\mathbf{A = USV^T}$

Here,

$\mathbf{A \rightarrow}$ **Given Matrix**

$\mathbf{U \rightarrow}$ **Matrix of Eigen Vectors of the Matrix** $\mathbf{AA^T}$

$\mathbf{V \rightarrow}$ **Matrix of Eigen Vectors of the Matrix** $\mathbf{A^TA}$

$\mathbf{S \rightarrow}$ **Roots of Eigen Values of the Matrix** $\mathbf{AA^T}$ **or** $\mathbf{A^TA}$

$\mathbf{V^T \rightarrow}$ **Transpose of the Matrix** $\mathbf{V}$

Example 68: **Compute Singular Value Decomposition (SVD) for the given matrix** $\begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

Answer: Let, $\mathbf{A} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

In order to find U, Now we need to find the eigenvectors for AA^T

Given, $\mathbf{A} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

$$\therefore \mathbf{A^T} = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

So, $\mathbf{AA^T} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 4+4 & 2-2 \\ 2-2 & 1+1 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} \quad \text{--------------------------------(i)}$$

Now we need to find the eigenvalues of $\mathbf{AA^T}$

$$\mathbf{AA^T - \lambda I} = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

$$= \begin{pmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{pmatrix}$$

$$\left|\mathbf{AA^T - \lambda I}\right| = 0$$

$$\Rightarrow \begin{vmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (8-\lambda)(2-\lambda) = 0$$

$$\Rightarrow (\lambda-8)(\lambda-2) = 0$$

$$\Rightarrow \lambda = 8,2$$

Therefore eigenvalues are 8 and 2. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$\boldsymbol{D} = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

The matrix of singular values $\boldsymbol{S} = \sqrt{\boldsymbol{D}} = \begin{pmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$

$$\mathbf{S} = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{---------------------------(ii)}$$

Now we need to find the eigenvector of $\mathbf{AA^T}$

We have, $\mathbf{Au_1} = \lambda \mathbf{u_1}$ $\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad$ $[\boldsymbol{AV} = \lambda \boldsymbol{V}]$

Let $\mathbf{u_1} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$

$\therefore \mathbf{AA^T u_1} = \lambda \mathbf{u_1}$

$\begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = 8\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$ $\quad\quad\quad\quad\quad\quad\quad\quad$ [For $\lambda = \mathbf{8}$ & from (i) putting the value of $\boldsymbol{AA^T}$]

$$\Rightarrow \begin{pmatrix} \mathbf{8x} \\ \mathbf{2y} \end{pmatrix} = \begin{pmatrix} \mathbf{8x} \\ \mathbf{8y} \end{pmatrix}$$

$$\Rightarrow \mathbf{8x = 8x}$$
$$\mathbf{2y = 8y}$$

$$\Rightarrow \mathbf{8x - 8x = 0}$$
$$\mathbf{2y - 8y = 0}$$

$$\Rightarrow \mathbf{8x = 8x}$$
$$\mathbf{-6y = 0}$$

$$\Rightarrow \mathbf{8x = 8x}$$
$$\mathbf{y = 0}$$

$\therefore \mathbf{y = 0}$

Let $\mathbf{x = -1}$

$\therefore \boldsymbol{u}_1 = \begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ It is a unit vector or normal vector. Since the length of $\mathbf{u_1}$ is $|\mathbf{u_1}| = \sqrt{(-1)^2 + (0)^2} = \sqrt{1} = 1$

Again, Let $\mathbf{u_2} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$

Here,

$\therefore \mathbf{AA^T u_2} = \lambda \mathbf{u_2}$

$\begin{pmatrix} \mathbf{8} & \mathbf{0} \\ \mathbf{0} & \mathbf{2} \end{pmatrix}\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \mathbf{2}\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$ [For $\lambda = \mathbf{2}$ & from (i) putting the value of $\boldsymbol{AA^T}$]

$\Rightarrow \begin{pmatrix} \mathbf{8x} \\ \mathbf{2y} \end{pmatrix} = \begin{pmatrix} \mathbf{2x} \\ \mathbf{2y} \end{pmatrix}$

$$\Rightarrow \mathbf{8x = 2x}$$
$$\mathbf{2y = 2y}$$

$$\Rightarrow \mathbf{8x - 2x = 0}$$
$$\mathbf{2y = 2y}$$

$$\Rightarrow \mathbf{6x = 0}$$
$$\mathbf{2y = 2y}$$

$\therefore \boldsymbol{x} = \mathbf{0}$

$\mathbf{2}\boldsymbol{y} = \mathbf{2}\boldsymbol{y}$

Let $\mathbf{y = 1}$

$\therefore \boldsymbol{u_2} = \begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{1} \end{pmatrix}$ It is a unit vector or normal vector. Since the length of $\mathbf{u_1}$ is $|\mathbf{u_2}| = \sqrt{(\mathbf{0})^2 + (\mathbf{1})^2} = \sqrt{\mathbf{1}} = \mathbf{1}$

$\therefore \boldsymbol{U} = \begin{pmatrix} \boldsymbol{u_1} & \boldsymbol{u_2} \end{pmatrix} = \begin{pmatrix} \mathbf{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{pmatrix}$ ---------------------------------(iii)

Now we need to find the eigenvectors for $A^T A$

Given, $\mathbf{A} = \begin{pmatrix} \mathbf{2} & \mathbf{-2} \\ \mathbf{1} & \mathbf{1} \end{pmatrix}$

$\therefore \mathbf{A^T} = \begin{pmatrix} \mathbf{2} & \mathbf{1} \\ \mathbf{-2} & \mathbf{1} \end{pmatrix}$

$\therefore \mathbf{A^T A} = \begin{pmatrix} \mathbf{2} & \mathbf{1} \\ \mathbf{-2} & \mathbf{1} \end{pmatrix}\begin{pmatrix} \mathbf{2} & \mathbf{-2} \\ \mathbf{1} & \mathbf{1} \end{pmatrix}$

$= \begin{pmatrix} \mathbf{5} & \mathbf{-3} \\ \mathbf{-3} & \mathbf{5} \end{pmatrix}$ -----------------------------------(iv)

Now we need to find the eigenvalues of $A^T A$

$$\therefore\ \mathbf{A^TA-\lambda I}=\begin{pmatrix}5 & -3\\ -3 & 5\end{pmatrix}-\lambda\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$$

$$=\begin{pmatrix}5 & -3\\ -3 & 5\end{pmatrix}-\begin{pmatrix}\lambda & 0\\ 0 & \lambda\end{pmatrix}$$

$$=\begin{pmatrix}5-\lambda & -3\\ -3 & 5-\lambda\end{pmatrix}$$

$$\therefore\ \left|\mathbf{A^TA-\lambda I}\right|=0$$

$$\Rightarrow\begin{vmatrix}5-\lambda & -3\\ -3 & 5-\lambda\end{vmatrix}=0$$

$$(5-\lambda)(5-\lambda)-9=0$$
$$\Rightarrow 25-10\lambda+\lambda^2-9=0$$
$$\Rightarrow \lambda^2-10\lambda+16=0$$
$$\Rightarrow (\lambda-8)(\lambda-2)=0$$
$$\Rightarrow \lambda=8,2$$

Therefore eigenvalues are 8 and 2. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$\boldsymbol{D}=\begin{pmatrix}8 & 0\\ 0 & 2\end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

The matrix of singular values $\boldsymbol{S}=\begin{pmatrix}\sqrt{8} & 0\\ 0 & \sqrt{2}\end{pmatrix}$

$$\mathbf{S}=\begin{pmatrix}2\sqrt{2} & 0\\ 0 & \sqrt{2}\end{pmatrix}\text{-----------------------------(v)}$$

Now we need find the eigenvectors of A^T A which are the columns of V.

We have, $\mathbf{Av_1=\lambda v_1}$ $\qquad\qquad\qquad\qquad$ $[AV=\lambda V]$

Let $\mathbf{v_1}=\begin{pmatrix}x\\ y\end{pmatrix}$

$$\therefore\ \mathbf{A^TAv_1=\lambda v_1}$$

$$\begin{pmatrix}5 & -3\\ -3 & 5\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\lambda\begin{pmatrix}x\\ y\end{pmatrix}$$

$$\begin{pmatrix}5 & -3\\ -3 & 5\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=8\begin{pmatrix}x\\ y\end{pmatrix}$$ [For $\lambda=\mathbf{8}$ & from (iv) putting the value of $\boldsymbol{A^TA}$]

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\begin{pmatrix} 5x-3y \\ -3x+5y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\Rightarrow 5x-3y = 8x$$
$$-3x+5y = 8y$$

$$\Rightarrow -3x-3y = 0$$
$$-3x-3y = 0$$

$$\Rightarrow x+y = 0$$
$$x+y = 0$$

$$\Rightarrow x+y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Let y = 1 $\therefore \mathbf{x = -1}$

$\therefore \boldsymbol{v_1} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$; It is not unit vector or normal vector. Since the length of $\mathbf{v_1}$ is $|\mathbf{v_1}| = \sqrt{(-1)^2 + (1)^2} = \sqrt{2}$; this vector $\mathbf{v_1}$ has to be converted into unit vector by dividing its magnitude or its length.

We have, $\mathbf{v_1} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$\therefore \mathbf{v_1} = \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ [Dividing by its magnitude or its length $\sqrt{2}$]

$\mathbf{v_1} = \begin{pmatrix} \frac{-\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$ [*Multiplying by* $\sqrt{2}$ *on numerator and denominator]*

Similarly,

We have, $\mathbf{Av_2} = \lambda \mathbf{v_2}$ $[AV = \lambda V]$

Let $\mathbf{v_2} = \begin{pmatrix} x \\ y \end{pmatrix}$

$\therefore \mathbf{A^T A v_2} = \lambda \mathbf{v_2}$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 2\begin{pmatrix} x \\ y \end{pmatrix}$$ [For $\lambda = 2$ & from (iv) putting the value of $A^T A$]

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\begin{pmatrix} 5x - 3y \\ -3x + 5y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\Rightarrow \begin{aligned} 3x - 3y &= 0 \\ -3x + 3y &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} x - y &= 0 \\ x - y &= 0 \end{aligned}$$

$$\Rightarrow x - y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Let y = 1 $\therefore x = 1$

$\therefore v_2 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ It is not unit vector or normal vector. Since the length of $\mathbf{v_2}$ is $|\mathbf{v_2}| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$; this vector $\mathbf{v_2}$ has to be converted into unit vector by dividing its magnitude or its length.

We have,

$$\mathbf{v_2} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Thus, $\mathbf{v_2} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ [Dividing by its magnitude or its length $\sqrt{2}$]

$\Rightarrow \mathbf{v_2} = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$ *[Multiplying by $\sqrt{2}$ on numerator and denominator]*

Now we need to construct the augmented orthogonal matrix V,

$$\mathbf{V} = (\mathbf{v_1} \quad \mathbf{v_2}) = \begin{pmatrix} \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\therefore V^T = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \qquad \text{------------------------------------(vi)}$$

From (ii), (iii) & (vi), Putting the values of U, S, V^T **in the relation A = USV^T**

Therefore **A = USV^T**

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix}\begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \textit{Answer}$$

Justification Test:

A = USV^T

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix}\begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \text{ [Putting the value of U , } V^T \text{, S]}$$

A = USV^T

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -2\sqrt{2} \times \frac{\sqrt{2}}{2} + 0 & 2\sqrt{2} \times \frac{\sqrt{2}}{2} + 0 \\ 0 + \sqrt{2} \times \frac{\sqrt{2}}{2} & 0 + \sqrt{2} \times \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -2 & 2 \\ 1 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} -1 \times (-2) + 0 & (-1) \times 2 + 0 \\ 0 + 1 & 0 + 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} \textit{ [Given Matrix]} \text{ Proved}$$

Example 69: Compute Singular Value Decomposition (SVD) for the given matrix

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

In order to find U, In order to find U, Now we need to find the eigenvectors for AA^T

Given,

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

So, $\mathbf{AA^T} = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix}$ ------------------------------(i)

Now we need find the eigenvalues of $\mathbf{AA^T}$

$$\left|\mathbf{AA^T} - \lambda \mathbf{I}\right| = 0$$

$$\left|\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right| = 0$$

$$\left|\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\right| = 0$$

Therefore,

$$\begin{vmatrix} 11-\lambda & 1-0 \\ 1-0 & 11-\lambda \end{vmatrix} = 0$$

$$(11-\lambda)(11-\lambda) - 1 = 0$$

$$\Rightarrow 121 - 11\lambda - 11\lambda + \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 22\lambda + 121 - 1 = 0$$

$$\Rightarrow \lambda^2 - 22\lambda + 120 = 0$$

$$\Rightarrow \lambda^2 - 10\lambda - 12\lambda + 120 = 0$$

$$\Rightarrow (\lambda - 10)(\lambda - 12) = 0$$

$$\Rightarrow \lambda = 10{,}12$$

Therefore eigenvalues are 10 and 12. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$\boldsymbol{D} = \begin{pmatrix} 12 & 0 \\ 0 & 10 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

The matrix of singular values $\boldsymbol{S} = \sqrt{\boldsymbol{D}} = \begin{pmatrix} \sqrt{12} & 0 \\ 0 & \sqrt{10} \end{pmatrix}$ ---------------------------(ii)

Now we need to find the eigenvector of AA^T

We have, $\mathbf{Au_1} = \lambda \mathbf{u_1}$ $[AV = \lambda V]$

Let $\mathbf{u_1} = \begin{pmatrix} x \\ y \end{pmatrix}$

$\therefore \mathbf{AA^T u_1} = \lambda \mathbf{u_1}$

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 12 \begin{pmatrix} x \\ y \end{pmatrix}$$ [For $\lambda = 12$ & from (i) putting the value of AA^T]

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 12x \\ 12y \end{pmatrix}$$

$$\begin{pmatrix} 11x + y \\ x + 11y \end{pmatrix} = \begin{pmatrix} 12x \\ 12y \end{pmatrix}$$

$$\Rightarrow \begin{aligned} 11x + y &= 12x \\ x + 11y &= 12y \end{aligned}$$

$$\Rightarrow \begin{aligned} -x + y &= 0 \\ x - y &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} x - y &= 0 \\ x - y &= 0 \end{aligned}$$

$$\Rightarrow x - y = 0$$

Let y = 1 $\therefore x = 1$

$\therefore u_1 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ It is not unit vector or normal vector. Since the length of $\mathbf{u_1}$ is $|\mathbf{u_1}| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$; this vector $\mathbf{u_1}$ has to be converted into unit vector by dividing its magnitude or its length.

Since U has an orthonormal basis, u_1 needs to be of length one. We divide $u_1$by its magnitude to accomplish this,

Thus, $\therefore \mathbf{u}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ [Dividing by its magnitude or its length $\sqrt{2}$]

Again,

We have, $Au_2 = \lambda u_2$ $[AV = \lambda V]$

$\therefore AA^T u_2 = \lambda u_2$

Let $\mathbf{u_2} = \begin{pmatrix} x \\ y \end{pmatrix}$

For $\lambda = 10$

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 10\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 10\begin{pmatrix} x \\ y \end{pmatrix}$$ [For $\lambda = 10$ & from (i) putting the value of AA^T]

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 10x \\ 10y \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{11x + y} \\ \mathbf{x + 11y} \end{pmatrix} = \begin{pmatrix} \mathbf{10x} \\ \mathbf{10y} \end{pmatrix}$$

$$\Rightarrow \mathbf{11x + y = 10x}$$
$$\mathbf{x + 11y = 10y}$$

$$\Rightarrow \quad \mathbf{x + y = 0}$$
$$\mathbf{x + y = 0}$$

$$\Rightarrow \mathbf{x + y = 0}$$
$$\mathbf{x + y = 0}$$

$$\Rightarrow \mathbf{x + y = 0}$$

Let y = -1 $\therefore \mathbf{x = 1}$

$\therefore \boldsymbol{u_2} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ It is not unit vector or normal vector. Since the length of $\mathbf{u_2}$ is $|\mathbf{u_2}| = \sqrt{(1)^2 + (-1)^2} = \sqrt{2}$; this vector $\mathbf{u_2}$ has to be converted into unit vector by dividing its magnitude or its length.

Since U has an orthonormal basis, u_2 needs to be of length one. We divide u_2 by its magnitude to accomplish this,

Thus, $\therefore \mathbf{u_2} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{pmatrix}$ [Dividing by its magnitude or its length $\sqrt{2}$]

Since $\mathbf{u_2}$ is already of length one, dividing by the magnitude we get the same vector back,

Setting up the augmented matrix U

$$\mathbf{U = (u_1 \quad u_2) =} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \qquad \text{------------------------------------(iii)}$$

Now we need to find the eigenvectors of $A^T A$ which are the columns of V.

Start with the matrix

$$\mathbf{A} = \begin{bmatrix} \mathbf{3} & \mathbf{1} & \mathbf{1} \\ \mathbf{-1} & \mathbf{3} & \mathbf{1} \end{bmatrix}$$

In order to find V, we have to start with $\mathbf{A^T A}$ the transpose of A is

$$\mathbf{A^T} = \begin{bmatrix} \mathbf{3} & \mathbf{-1} \\ \mathbf{1} & \mathbf{3} \\ \mathbf{1} & \mathbf{1} \end{bmatrix}$$

$$\therefore \mathbf{A^T A} = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

$$= \begin{pmatrix} 3\times 3+1 & 3-3 & 3-1 \\ 3-3 & 1+9 & 1+3 \\ 3-1 & 1+3 & 1+1 \end{pmatrix}$$

$$= \begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \qquad \text{---(iv)}$$

Now we need to find the eigenvalue of $\mathbf{A^T A}$

$$\left|\mathbf{A^T A} - \lambda \mathbf{I}\right| = \mathbf{0}$$

$$\left| \begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right| = 0$$

$$\left| \begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = 0$$

Therefore,

$$\left| \begin{pmatrix} 10-\lambda & 0 & 2 \\ 0 & 10-\lambda & 4 \\ 2 & 4 & 2-\lambda \end{pmatrix} \right| = 0$$

$$\begin{vmatrix} 10-\lambda & 0 & 2 \\ 0 & 10-\lambda & 4 \\ 2 & 4 & 2-\lambda \end{vmatrix} = 0$$

$$(10-\lambda)\{(10-\lambda)(2-\lambda)-16\} + 2\{0-2(10-\lambda)\} = 0$$

$$\Rightarrow (10-\lambda)(20-10\lambda-2\lambda+\lambda^2-16) - 4(10-\lambda) = 0$$

$$\Rightarrow (10-\lambda)(20-10\lambda-2\lambda+\lambda^2-16-4) = 0$$

$$\Rightarrow (10-\lambda)(\lambda^2-12\lambda) = 0$$

$$\Rightarrow \lambda(10-\lambda)(\lambda-12) = 0$$

$$\Rightarrow \lambda = 0, 10, 12$$

Therefore eigenvalues are 0, 10 and 12. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$\boldsymbol{D} = \begin{pmatrix} 12 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives, The matrix of singular values $S = \sqrt{D} = \begin{pmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \\ 0 & 0 & 0 \end{pmatrix}$ -------------------------------(v)

Now we need to find the eigenvector of A^TA which, are the columns of V.

We have, $Av_1 = \lambda v_1$ $\qquad$ $[AV = \lambda V]$

$\therefore A^T A v_1 = \lambda v_1$

Let $\mathbf{v_1} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix}$

$\mathbf{A^T A v_1 = \lambda v_1}$

$\begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 12 \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ $\qquad$ [For $\lambda = \mathbf{12}$ & from (iv) putting the value of $A^T A$]

$\begin{pmatrix} 10x + 0 + 2z \\ 0 + 10y + 4z \\ 2x + 4y + 2z \end{pmatrix} = \begin{pmatrix} 12x \\ 12y \\ 12z \end{pmatrix}$

$$\Rightarrow 10x + 2z = 12x$$
$$10y + 4z = 12y$$
$$2x + 4y + 2z = 12z$$

$$\Rightarrow 2x + 4y + 2z = 12z$$
$$10x + 2z = 12x$$
$$10y + 4z = 12y$$

$$\Rightarrow 2x + 4y - 10z = 0$$
$$-2x + 2z = 0$$
$$-2y + 4z = 0$$

$$\Rightarrow 2x + 4y - 10z = 0$$
$$-2x + 0.y + 2z = 0$$
$$-2y + 4z = 0$$

We have,

$\mathbf{L_i \rightarrow -a_{i1}L_1 + a_{11}L_i}$

i = 2,

$\mathbf{L_2 \rightarrow -a_{21}L_1 + a_{11}L_2}$

$\rightarrow$ -(-2)(2x + 4y-10z)+2(-2x + 0.y + 2z)

$\rightarrow$ 4x + 8y - 20z - 4x + 0 + 4z

$\rightarrow$ 8y -16z

Thus we obtain a new system,

$$\begin{aligned} 2x + 4y - 10z &= 0 \\ 8y - 16z &= 0 \\ -2y + 4z &= 0 \end{aligned}$$

$$\Rightarrow \quad \begin{aligned} 2x + 4y - 10z &= 0 \\ y - 2z &= 0 \\ y - 2z &= 0 \end{aligned}$$

Since the 1st and 2nd equations are identical. We can disregard one of them

$$\begin{aligned} 2x + 4y - 10z &= 0 \\ y - 2z &= 0 \end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular (3-2) =1 free variable, which is z.

Here free variable is z, Let z = 1, $\therefore$ $\mathbf{y = 2}$**, then** $\mathbf{2x + 4 \times 2 - 10 \times 1 = 0}$

$$\therefore \mathbf{x = 1}$$

$$\therefore v_1 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

$$|\mathbf{v_1}| = \sqrt{(1)^2 + (2)^2 + (1)^2} = \sqrt{6}$$

Since V has an orthonormal basis, v_1 needs to be of length one. We divide v_1 by it's magnitude to accomplish this, Thus, $\therefore \mathbf{v_1} = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix}$

Again,

We have, $Av_2 = \lambda v_2$ $\qquad\qquad\qquad [AV = \lambda V]$

$$\therefore A^T A v_2 = \lambda v_2$$

Let $\mathbf{v_2} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix}$

$$\mathbf{A^T A v_2 = \lambda v_2}$$

$$\begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 10 \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$ [For $\lambda = \mathbf{10}$ & from (iv) putting the value of $A^T A$]

$$\begin{pmatrix} 10x + 0 + 2z \\ 0 + 10y + 4z \\ 2x + 4y + 2z \end{pmatrix} = \begin{pmatrix} 10x \\ 10y \\ 10z \end{pmatrix}$$

$$\Rightarrow \mathbf{10x + 2z = 10x}$$
$$\mathbf{10y + 4z = 10y}$$
$$\mathbf{2x + 4y + 2z = 10z}$$

$$\Rightarrow \mathbf{2x + 4y + 2z = 10z}$$
$$\mathbf{10x + 2z = 10x}$$
$$\mathbf{10y + 4z = 10y}$$

$$\Rightarrow \mathbf{2x + 4y - 8z = 0}$$
$$\mathbf{2z = 0}$$
$$\mathbf{4z = 0}$$

$$\Rightarrow \mathbf{2x + 4y - 8z = 0}$$
$$\mathbf{2z = 0}$$
$$\mathbf{2z = 0}$$

$$\Rightarrow \mathbf{2x + 4y - 8z = 0}$$
$$\mathbf{2z = 0}$$

$$\Rightarrow \mathbf{2x + 4y - 8z = 0}$$
$$\mathbf{z = 0}$$

$$\therefore \mathbf{z = 0}$$

Then $\Rightarrow \mathbf{2x + 4y - 8 \times 0 = 0}$

$$\Rightarrow \mathbf{2x + 4y = 0}$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Here free variable is y, Let y = -1, then x = 2

$$\therefore v_2 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \mathbf{2} \\ \mathbf{-1} \\ \mathbf{0} \end{pmatrix}$$

$$|\mathbf{v_2}| = \sqrt{\mathbf{(2)^2 + (-1)^2 + (0)^2}} = \sqrt{\mathbf{5}}$$

Since V has an orthonormal basis, ν_2 needs to be of length one. We divide ν_2 by it's magnitude to accomplish this, Thus, $\therefore \mathbf{v_2} = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{-1}{\sqrt{5}} \\ 0 \end{pmatrix}$

Again,

We have, $\boldsymbol{Av_3} = \lambda \boldsymbol{v_3}$ $\qquad$ $[\boldsymbol{AV} = \lambda \boldsymbol{V}]$

We have, $\mathbf{A^T A v_3} = \lambda \mathbf{v_3}$

Let $\mathbf{v_3} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix}$

$$\mathbf{A^T A v_3} = \lambda \mathbf{v_3}$$

$$\begin{pmatrix} \mathbf{10} & \mathbf{0} & \mathbf{2} \\ \mathbf{0} & \mathbf{10} & \mathbf{4} \\ \mathbf{2} & \mathbf{4} & \mathbf{2} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix} = 0 \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix}$$

[For $\lambda = \mathbf{0}$ & from (iv) putting the value of $\boldsymbol{A^T A}$]

$$\begin{pmatrix} \mathbf{10x + 0 + 2z} \\ \mathbf{0 + 10y + 4z} \\ \mathbf{2x + 4y + 2z} \end{pmatrix} = \begin{pmatrix} \mathbf{0x} \\ \mathbf{0y} \\ \mathbf{0z} \end{pmatrix}$$

$$\Rightarrow \mathbf{10x + 2z = 0}$$
$$\mathbf{10y + 4z = 0}$$
$$\mathbf{2x + 4y + 2z = 0}$$
$$\Rightarrow \mathbf{2x + 4y + 2z = 0}$$
$$\mathbf{10x + 2z = 0}$$
$$\mathbf{10y + 4z = 0}$$
$$\Rightarrow \mathbf{2x + 4y + 2z = 0}$$
$$\mathbf{10x + 0.y + 2z = 0}$$
$$\mathbf{10y + 4z = 0}$$

$$\mathbf{L_i \rightarrow -a_{i1}L_1 + a_{11}L_i}$$

i = 2,

$$\mathbf{L_2 \rightarrow -a_{21}L_1 + a_{11}L_2}$$

$\rightarrow$ -10(2x + 4y+2z)+2(10x + 0.y + 2z)

$\rightarrow$ -20x - 40y - 20z +20x + 0 + 4z

$\rightarrow$ -40y -16z

Thus we obtain the following system:

$$\Rightarrow 2x + 4y + 2z = 0$$
$$-40y - 16z = 0$$
$$10y + 4z = 0$$

$$\Rightarrow 2x + 4y + 2z = 0$$
$$10y + 4z = 0$$
$$10y + 4z = 0$$

$$\Rightarrow 2x + 4y + 2z = 0$$
$$10y + 4z = 0$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular (3-2) =1 free variable, which is z

Here free variable is z, Let z = -5, then y = 2, then x =1

$$\therefore v_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -5 \end{pmatrix}$$

$$|v_3| = \sqrt{(1)^2 + (2)^2 + (-5)^2} = \sqrt{30}$$

Since V has an orthonormal basis, v_3 needs to be of length one. We divide v_3 by it's magnitude to accomplish this, Thus, $\therefore v_3 = \begin{pmatrix} \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{30}} \\ \frac{-5}{\sqrt{30}} \end{pmatrix}$

Now we need to construct the augmented orthogonal matrix V,

$$V = (v_1 \quad v_2 \quad v_3)$$

All this to give us

$$V = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{6}} & \frac{-1}{\sqrt{5}} & \frac{2}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} & 0 & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

when we really want its transpose

$$V^T = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

For S we take the square roots of the non-zero eigenvalues and populate the diagonal with them, putting the largest in s_{11}, the next largest in s_{22} and so on until the smallest value ends up in s_{mn}. The non-zero eigenvalues of U and V are always the same, so that's why it doesn't matter which one we take them from. Because we are doing full SVD, instead of reduced SVD (next section), we have to add a zero column vector to S so that it is of the proper dimensions to allow multiplication between U and V. The diagonal entries in S are the singular values of A, the columns in U are called left singular vectors, and the columns in V are called right singular vectors.

$$S = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow S = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix}$ [Ignoring third row of the matrix as all elements are zero in the last row]

Now we have all the pieces of the puzzle

$A = USV^T$

Putting the value of U, V^T, S in $A = USV^T$

$A = USV^T$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sqrt{12}}{\sqrt{2}} & \frac{\sqrt{10}}{\sqrt{2}} & 0 \\ \frac{\sqrt{12}}{\sqrt{2}} & \frac{-\sqrt{10}}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

$$= \begin{bmatrix} \mathbf{3} & \mathbf{1} & \mathbf{1} \\ \mathbf{-1} & \mathbf{3} & \mathbf{1} \end{bmatrix} \text{[Proved]}$$

.

Home Task:

Find the singular value decomposition of the matrix

i. $A = \begin{bmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{0} & \mathbf{1} \\ \mathbf{1} & \mathbf{0} \end{bmatrix}$ ii. $A = \begin{bmatrix} \mathbf{2} & \mathbf{1} & \mathbf{-2} \end{bmatrix}$

iii. $A = \begin{bmatrix} \mathbf{2} & \mathbf{4} \\ \mathbf{1} & \mathbf{3} \\ \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$

Hints:

$$U = \begin{bmatrix} 0.82 & -0.58 & 0 & 0 \\ 0.58 & 0.82 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$V = \begin{bmatrix} 0.40 & -0.91 \\ 0.91 & 0.40 \end{bmatrix}$$

$$S = \begin{bmatrix} 5.47 & 0 \\ 0 & 0.37 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Singular Value Decomposition:

The Singular Value Decomposition is an important factorization of a real or complex matrix, with several applications in signal processing and statistics. Applications, which employ the SVD, include computing the pseudo-inverse, matrix approximation, and determining the rank, range and null space of a matrix.

The spectral theorem says that normal matrices, which by definition must be square, can be unitarily diagonalized using a basis of Eigen vectors. The SVD can be seen as a generalization of the spectral theorem to arbitrary, not necessarily square, matrices.

Algorithm of Singular Value Decomposition:

To find the Singular Value Decomposition of the matrix:

1. Find the eigenvalues of the matrix $\mathbf{A^T A}$ and arrange them in descending order
2. Find the number of nonzero eigenvalues of the matrix $\mathbf{A^T A}$
3. Find the orthogonal eigenvectors of the matrix $\mathbf{A^T A}$ corresponding to the obtained eigenvalues, and arrange them in the same order to form the column-vectors of the matrix V
4. Form a diagonal matrix $\mathbf{S} = \sum = \sigma$ placing on the leading diagonal of it the square roots $\sigma_i = \sqrt{\lambda_i}$ of $\mathbf{p = min\{m,n\}}$ first eigenvalues of the matrix $\mathbf{A^T A}$ got in I in descending order
5. Find the first column-vectors of the matrix U, $\mathbf{u_i = \sigma_i^{-1} A v_i (i = 1 : r)}$
6. Add to the matrix U the rest of m-r vectors using the Gram-Schmidt Orthogonalization process

Image Processing with SVD

We want to approximate the $\mathbf{m \times n}$ matrix A by using far fewer entries than in the original matrix.

$$\mathbf{A = USV^T = SUV^T}$$

$$\mathbf{\Rightarrow A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \ldots\ldots\ldots\ldots\ldots\ldots + \sigma_r u_r v_r^T + 0.u_{r+1} v_{r+1}^T + \ldots\ldots\ldots}$$

Since the singular values are always greater than zero. Adding on the dependent terms where the singular values are equal to zero does not effect the image. The terms at the end of the equation zero out leaving us with:

$$\mathbf{\Rightarrow A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \ldots\ldots\ldots\ldots\ldots\ldots + \sigma_r u_r v_r^T}$$

Conclusion: We can further approximate the matrix (original image) by leaving off more singular terms of the matrix A. Since the singular values are arranged in descending order, the last terms will have the least effect on the overall image.

Doing this (leaving more singular values) reduces the amount of space required to store the image on a computer.

Justification using above example:

The original image is: $\mathbf{A} = \begin{pmatrix} \mathbf{2} & \mathbf{-2} \\ \mathbf{1} & \mathbf{1} \end{pmatrix}$

$\mathbf{S} = \begin{pmatrix} \mathbf{2\sqrt{2}} & \mathbf{0} \\ \mathbf{0} & \mathbf{\sqrt{2}} \end{pmatrix}$ **i.e.**$\mathbf{\sigma_1 = 2\sqrt{2}, \quad \sigma_2 = \sqrt{2}}$

$U = (u_1 \ \ u_2) = \begin{pmatrix} \mathbf{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{pmatrix}$

$V = (v_1 \ \ v_2) = \begin{pmatrix} \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$

$$\Rightarrow A = \sigma_1 u_1 v_1{}^T + \sigma_2 u_2 v_2{}^T$$

$$= 2\sqrt{2}\begin{pmatrix} -1 \\ 0 \end{pmatrix}\begin{pmatrix} \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} + \sqrt{2}\begin{pmatrix} 0 \\ 1 \end{pmatrix}\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$= \begin{pmatrix} -2\sqrt{2} \\ 0 \end{pmatrix}\begin{pmatrix} \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} + \begin{pmatrix} 0 \\ \sqrt{2} \end{pmatrix}\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$= \begin{pmatrix} 2 & -2 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} = A \text{ (Original Image can be obtained)}$$

We can get approximate image by using less singular values. Say number of singular values are 10 i.e. $\sigma_1, \sigma_2, \ldots\ldots\ldots\ldots \sigma_{10}$.

Then,

$$\Rightarrow A = \sigma_1 u_1 v_1{}^T + \sigma_2 u_2 v_2{}^T + \ldots\ldots\ldots\ldots + \sigma_{10} u_{10} v_{10}{}^T$$

If we use 5 singular values then

$$\Rightarrow A = \sigma_1 u_1 v_1{}^T + \sigma_2 u_2 v_2{}^T + \ldots\ldots\ldots\ldots + \sigma_5 u_5 v_5{}^T$$

In this result we can get apprximate image of the original image (A)

Since large eigenvalues carry significant representation of the matrix, we arrange the values of the Eigen Values from the largest to the smallest and drop/ignore the values towards the last since small eigenvalues are less indicator.

What is the use of SVD?

01. SVD can be used to compute optimal low-rank approximations of arbitrary matrices.
02. Face recognition: represent the face images as eigenfaces and compute distance between the query face image in the principal component space.

B. LU Decomposition

A procedure for decomposing a $N \times N$ matrix A into a product of a lower triangular matrix L and an upper triangular matrix U,

$$A = LU$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix}\begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

LU decomposition is implemented in *Mathematica* as LU Decomposition[*m*].

Example 70

Solve the linear system of equations using LU decomposition

$$u_1 + 2u_2 + 3u_3 = 5$$
$$2u_1 - 4u_2 + 6u_3 = 18$$
$$3u_1 - 9u_2 - 3u_3 = 6$$

Solution:

Given the linear system of equations

$$u_1 + 2u_2 + 3u_3 = 5$$
$$2u_1 - 4u_2 + 6u_3 = 18$$
$$3u_1 - 9u_2 - 3u_3 = 6$$

This can be written as

$$\mathbf{Au = b} \qquad \text{------------------------------(i)}$$

Where,

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

Again, According to LU decomposition,

$$\mathbf{A = LU} \qquad \text{------------------------------(ii)}$$

Where,

$$\mathbf{L} = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \quad \text{and} \quad \mathbf{U} = \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

From (ii), $\mathbf{A = LU}$

$$\Rightarrow \mathbf{LU = A}$$

$$\begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}$$

$$\begin{pmatrix} l_{11}+0+0 & l_{11}u_{12}+0+0 & l_{11}u_{13}+0+0 \\ l_{21}+0+0 & l_{21}u_{12}+l_{22}+0 & l_{21}u_{13}+l_{22}u_{23}+0 \\ l_{31}+0+0 & l_{31}u_{12}+l_{32}+0 & l_{31}u_{13}+l_{32}u_{23}+l_{33} \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}$$

Now,

We can write

$l_{11} = 1$ ------------------------------------(iii)

$l_{11}u_{12} = 2$ ------------------------------------(iv)

$l_{11}u_{13} = 3$ ------------------------------------(v)

$l_{21} = 2$ ------------------------------------(vi)

$l_{21}u_{12} + l_{22} = -4$ ------------------------------------(vii)

$l_{21}u_{13} + l_{22}u_{23} = 6$ ------------------------------------(viii)

$l_{31} = 3$ ------------------------------------(ix)

$l_{31}u_{12} + l_{32} = -9$ ------------------------------------(x)

$l_{31}u_{13} + l_{32}u_{23} + l_{33} = -3$ ------------------------------------(xi)

From (iii) & (iv)

$l_{11} = 1$

$l_{11}u_{12} = 2$

$1.u_{12} = 2$

$u_{12} = 2$ ------------------------------------(xii)

From (iii) & (v)

$l_{11}u_{13} = 3$

$1.u_{13} = 3$

$u_{13} = 3$ ------------------------------------(xiii)

From (vii),

$l_{21}u_{12} + l_{22} = -4$

$2.2 + l_{22} = -4$ $[u_{12} = 2; l_{21} = 2]$

$4 + l_{22} = -4$

$\therefore l_{22} = -8$ ------------------------------------(xiv)

From (viii),

$$\mathbf{l_{21}u_{13} + l_{22}u_{23} = 6}$$

$$\mathbf{2.3 - 8u_{23} = 6} \qquad [\mathbf{l_{21} = 2\,;u_{13} = 3\,;\therefore l_{22} = -8}]$$

$$\mathbf{-8u_{23} = 6 - 6}$$

$$\mathbf{u_{23} = 0} \qquad \text{-------------------------------------(xv)}$$

From (x),

$$\mathbf{l_{31}u_{12} + l_{32} = -9}$$

$$\mathbf{3.2 + l_{32} = -9} \qquad [\mathbf{l_{31} = 3\,;u_{12} = 2}]$$

$$\mathbf{l_{32} = -9 - 6}$$

$$\mathbf{l_{32} = -15} \qquad \text{-------------------------------------(xvi)}$$

From (xi)

$$\mathbf{l_{31}u_{13} + l_{32}u_{23} + l_{33} = -3}$$

$$\mathbf{3.3 - 15.0 + l_{33} = -3} \quad [\mathbf{l_{31} = 3\,;u_{13} = 3\,;l_{32} = -15\,;u_{23} = 0}]$$

$$\mathbf{l_{33} = -3 - 9}$$

$$\mathbf{l_{33} = -12} \qquad \text{-------------------------------------(xvii)}$$

$$\therefore \mathbf{L} = \begin{pmatrix} \mathbf{l_{11}} & \mathbf{0} & \mathbf{0} \\ \mathbf{l_{21}} & \mathbf{l_{22}} & \mathbf{0} \\ \mathbf{l_{31}} & \mathbf{l_{32}} & \mathbf{l_{33}} \end{pmatrix}$$

$$\Rightarrow \mathbf{L} = \begin{pmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{2} & \mathbf{-8} & \mathbf{0} \\ \mathbf{3} & \mathbf{-15} & \mathbf{-12} \end{pmatrix}$$

And

$$\mathbf{U} = \begin{pmatrix} \mathbf{1} & \mathbf{u_{12}} & \mathbf{u_{13}} \\ \mathbf{0} & \mathbf{l} & \mathbf{u_{23}} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{pmatrix}$$

$$\mathbf{U} = \begin{pmatrix} \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{0} & \mathbf{l} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{pmatrix}$$

From (i),

$\mathbf{Au = b}$

$\mathbf{LUu = b}$ [from (ii), A = LU]

$\mathbf{LUu = b}$ -------------------------------------(xviii)

Let,

$\mathbf{Uu = v}$ -------------------------------------(xix)

Where,

Where, $\mathbf{v} = \begin{pmatrix} \mathbf{v_1} \\ \mathbf{v_2} \\ \mathbf{v_3} \end{pmatrix}$

From (xviii),

$\mathbf{LUu = b}$

$\mathbf{Lv = b}$

$$\Rightarrow \begin{pmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{2} & \mathbf{-8} & \mathbf{0} \\ \mathbf{3} & \mathbf{-15} & \mathbf{-12} \end{pmatrix} \begin{pmatrix} \mathbf{v_1} \\ \mathbf{v_2} \\ \mathbf{v_3} \end{pmatrix} = \begin{pmatrix} \mathbf{5} \\ \mathbf{18} \\ \mathbf{6} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \mathbf{1 \times v_1 + 0 \times v_2 + 0 \times v_3} \\ \mathbf{2 \times v_1 - 8 \times v_2 + 0 \times v_3} \\ \mathbf{3 \times v_1 - 15 \times v_2 - 12 \times v_3} \end{pmatrix} = \begin{pmatrix} \mathbf{5} \\ \mathbf{18} \\ \mathbf{6} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \mathbf{v_1} \\ \mathbf{2v_1 - 8v_2} \\ \mathbf{3v_1 - 15v_2 - 12v_3} \end{pmatrix} = \begin{pmatrix} \mathbf{5} \\ \mathbf{18} \\ \mathbf{6} \end{pmatrix}$$

$\mathbf{v_1 = 5}$ -------------------------------------(xx)

$\mathbf{2v_1 - 8v_2 = 18}$ -------------------------------------(xxi)

$\mathbf{3v_1 - 15v_2 - 12v_3 = 6}$ -------------------------------------(xxii)

From (xxi),

$\mathbf{2v_1 - 8v_2 = 18}$

$\mathbf{2.5 - 8v_2 = 18}$ [$\mathbf{v_1 = 5}$]

$\mathbf{10 - 8v_2 = 18}$

$\mathbf{-8v_2 = 18 - 10}$

$-8v_2 = 8$

$\therefore v_2 = -1$ --------------------------------------(xxiii)

From (xxii),

$3v_1 - 15v_2 - 12v_3 = 6$

$3.5 - 15(-1) - 12v_3 = 6$ $[\, v_1 = 5\,;\, v_2 = -1\,]$

$15 + 15 - 12v_3 = 6$

$30 - 12v_3 = 6$

$-12v_3 = 6 - 30$

$-12v_3 = -24$

$\therefore v_3 = 2$ --------------------------------------(xxiv)

From (xix),

$Uu = v$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1\times u_1 + 2\times u_2 + 3\times u_3 \\ 0\times u_1 + 1\times u_2 + 0\times u_3 \\ 0\times u_1 + 0\times u_2 + 1\times u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} u_1 + 2u_2 + 3u_3 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$u_1 + 2u_2 + 3u_3 = 5$ --------------------------------------(xxv)

$u_2 = -1$ --------------------------------------(xxvi)

$u_3 = 2$ --------------------------------------(xxvii)

From (xxv),

$$\mathbf{u_1 + 2u_2 + 3u_3 = 5}$$

$$\mathbf{u_1 + 2(-1) + 3.2 = 5}$$

$$\mathbf{u_1 - 2 + 6 = 5}$$

$$\mathbf{u_1 = 5 - 4}$$

$$\mathbf{u_1 = 1}$$

$\mathbf{u_1 = 1}$, $\mathbf{u_2 = -1}$, $\mathbf{u_3 = 2}$ Answer

Example 71: Given the following system of linear equations, determine the value of each of the variables using the LU decomposition method.

$$6x_1 - 2x_2 = 14$$
$$9x_1 - x_2 + x_3 = 21$$
$$3x_1 - 7x_2 + 5x_3 = 9$$